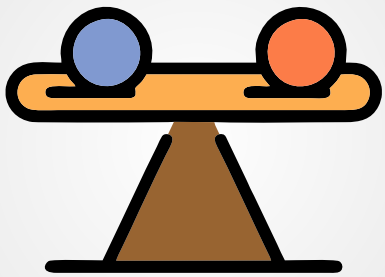
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Center of Mass, Momentum & Collision

(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Centre of mass

1. Find position vectors of mass center of a system of three particle of masses 1 kg, 2 kg and 3 kg located at position vectors $\vec{r}_1 = (4\hat{i} + 2\hat{j} - 3\hat{k})$ m, $\vec{r}_2 = (\hat{i} - 4\hat{j} + 2\hat{k})$ m and $\vec{r}_3 = (2\hat{i} - 2\hat{j} + \hat{k})$ m respectively.

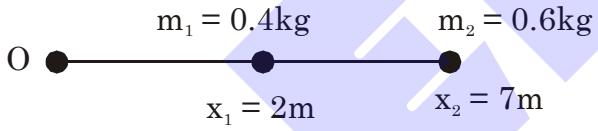
(A) $2\hat{i} - 2\hat{j} + \frac{2}{3}\hat{k}$ (B) $\hat{i} - \hat{j} + \frac{2}{3}\hat{k}$ (C) $2\hat{i} + 2\hat{j} - \frac{2}{3}\hat{k}$ (D) $4\hat{i} - 4\hat{j} + \frac{4}{3}\hat{k}$

CM0152

2. Two point mass 36 g and 72 g are located at coordinates (2 cm, 0) and (5 cm, 0). The coordinate of centre of mass will be at **CM0153**

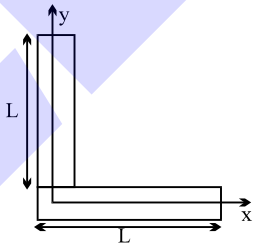
(A) (4, 0) (B) (4.5, 0) (C) (3.5, 0) (D) (3, 0)

3. A system consists of two masses connected by a massless rod lies along x–axis. The distance of centre of mass from O is **CM0154**



(A) 2m (B) 3m (C) 5m (D) 7m

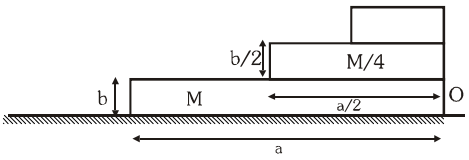
4. Centre of mass of two thin uniform rods of same length but made up of different materials & kept as shown, can be, if the meeting point is the origin of co-ordinates



(A) (L/2, L/2) (B) (2L/3, L/2) (C) (L/3, L/3) (D) (L/3, L/6)

CM0037

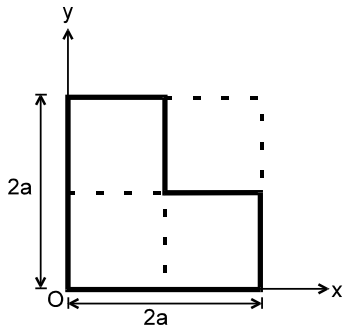
5. An infinite number of bricks are placed one over the other as shown in the figure. Each succeeding brick having half the length and breadth of its preceding brick and the mass of each succeeding bricks being 1/4th of the preceding one, take 'O' as the origin, the x-coordinate of centre of mass of the system of bricks is:- **CM0155**



(A) $-\frac{a}{7}$ (B) $\frac{3a}{7}$ (C) $-\frac{3a}{7}$ (D) $-\frac{2a}{7}$

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6. Find the position of centre of mass of the uniform planner section shown in figure with respect to the origin (O) **CM0156**



(A) (5a/6, 5a/6) (B) (5a/3, 5a/3) (C) (3a/4, 3a/4) (D) (5a/4, 5a/4)

7. A hemisphere and a solid cone of same density have a common base. The centre of mass of the common structure coincides with the centre of the common base. If R is the radius of hemisphere and h is height of the cone, then : **CM0157**

(A) $\frac{h}{R}=\sqrt{3}$ (B) $\frac{h}{R}=\frac{1}{\sqrt{3}}$ (C) $\frac{h}{R}=3$ (D) $\frac{h}{R}=\frac{1}{3}$

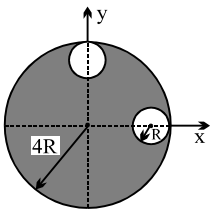
8. A uniform disc of radius R is put over another uniform disc of radius 2R of the same thickness and density. The peripheries of the two discs touch each other. The location of the centre of mass of the system from the centre of large disc :- **CM0158**

(A) At R/3 from the centre of the bigger disc towards the centre of the smaller disk.
(B) At R/3 from the centre of the bigger disc away from the centre of the smaller disk.
(C) At R/5 from the centre of the bigger disc away from the centre of the smaller disk.
(D) At R/5 from the centre of the bigger disc towards the centre of the smaller disk.

9. A thick straight wire of length πm is fixed at its midpoint and then bent in the form of a circle. The shift in its centre of mass is **CM0159**

(A) πm (B) 0.5 m (C) 2 m (D) $\frac{\pi}{2}$m

10. From the circular disc of radius 4R two small disc of radius R are cut off. The centre of mass of the new structure will be :



(A) $\hat{i}\frac{R}{5}+\hat{j}\frac{R}{5}$ (B) $-\hat{i}\frac{R}{5}+\hat{j}\frac{R}{5}$ (C) $\frac{-3R}{14}(\hat{i}+\hat{j})$ (D) None of these

CM0038

11. The centre of mass of a non uniform rod of length L whose mass per unit length $\lambda=\frac{Kx^2}{L}$,,Where k is a constant and x is the distance from one end is : **CM0160**

(A) $\frac{3L}{4}$ (B) $\frac{L}{8}$ (C) $\frac{K}{L}$ (D) $\frac{3K}{L}$

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Motion of CM and CM frame of reference

12. A man weighing 80 kg is standing at the centre of a flat boat and he is 20 m from the shore. He walks 8 m on the boat towards the shore and then halts. The boat weight 200 kg. How far is he from the shore at the end of this time ?

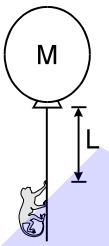
(A) 11.2 m (B) 13.8 m (C) 14.3 m (D) 15.4 m

CM0040

13. Mr. Verma (50 kg) and Mr. Mathur (60 kg) are sitting at the two extremes of a 4 m long boat (40 kg) standing still in water. To discuss a mechanics problem, they come to the middle of the boat. Neglecting friction with water, how far does the boat move in the water during the process ?

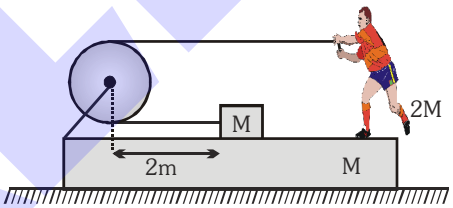
(A) 40/3 cm (B) 50/3 cm (C) 110/3 cm (D) 10 cm **CM0161**

14. The balloon, the light rope and the monkey shown in figure are at rest in the air. If the monkey reaches the top of the rope, by what distance does the balloon descend? Mass of the balloon = M, mass of the monkey = m and the length of the rope ascended by the monkey = L. **CM0162**



(A) ML/ (m + M) (B) mL/ (m + M) (C) L (D) 0

15. A block of mass M is tied to one end of a massless rope. The other end of the rope is in the hands of a man of mass 2M as shown in the figure. The block and the man are resting on a rough wedge of mass M as shown in the figure. The whole system is resting on a smooth horizontal surface. The man pulls the rope. Pulley is massless and frictionless. What is the displacement of the wedge when the block meets the pulley. (Man does not leave his position during the pull) **CM0163**



(A) 0.5m (B) 1m (C) Zero (D) 2/3 m

16. Two objects of masses 200 gm and 500 gm possess velocities $10\hat{i}$ m/s and $(3\hat{i}+5\hat{j})$ m/s respectively. The velocity of their centre of mass in m/s is– **CM0164**

(A) $5\hat{i}-25\hat{j}$ (B) $\frac{5}{7}\hat{i}-25\hat{j}$ (C) $5\hat{i}+\frac{25}{7}\hat{j}$ (D) $5\hat{i}-\frac{5}{7}\hat{j}$

17. A 2.0 kg particle has a velocity of $\vec{v}_1=(2.0\hat{i}-3.0\hat{j})$ m/s, and a 3.0 kg particle has a velocity $\vec{v}_2=(1.0\hat{i}+6.0\hat{j})$ m/s. velocities of both the particles in centroidal frame are **CM0165**

(A) $3.4\hat{i}-7.8\hat{j}$ & $(2.4\hat{i}+2.6\hat{j})$ (B) $1.6\hat{i}-3.4\hat{j}$ & $-(0.4\hat{i}+2.6\hat{j})$

(C) $0.6\hat{i}-5.4\hat{j}$ & $-(0.4\hat{i}+2.6\hat{j})$ (D) $0.3\hat{i}-0.4\hat{j}$ & $-(0.2\hat{i}+2.6\hat{j})$

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18. The figure shows the positions and velocities of two particles. If the particles move under the mutual attraction of each other, then the position of centre of mass at t =1 s is **CM0166**



(A) x=5m (B) x=6m (C) x=3m (D) x=2m

19. Two bodies of masses m_1 and m_2 are moving with velocity v_1 and v_2 respectively in the same direction. The total momentum of the system in the frame of reference attached to the centre of mass is (v is relative velocity between the masses) **CM0167**

(A) $\frac{m_1 m_2 v}{m_1 - m_2}$ (B) $\frac{2m_1 m_2 v}{m_1 + m_2}$ (C) Zero (D) $\frac{4m_1 m_2 v}{m_1 + m_2}$

20. Velocity of centre of mass of two bodies is v and the sum of the masses of two bodies is m. Kinetic energy of the system is **CM0168**

(A) will be equal to $1/2mv^2$ (B) will always be less than $1/2\ mv^2$

(C) will be greater than or equal to $1/2\ mv^2$ (D) will always be greater than $1/2\ mv^2$

21. **Statement-1 :** Two particles of mass 1 kg and 3 kg move towards each other under their mutual force of attraction. No other force acts on them. When the relative velocity of approach of the two particles is 2 m/s, their centre of mass has a velocity of 0.5 m/s. When the relative velocity of approach becomes 3 m/s, the velocity of the centre of mass is 0.75 m/s. **CM0169**

Statement-2 : The total kinetic energy as seen from ground is $\frac{1}{2}\mu v_{rel}^2 + \frac{1}{2}mv_c^2$ and in absence of external force, total energy remains conserved.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

22. Seven identical birds are flying south together at constant velocity. A hunter shoots one of them, which immediately dies and falls to the ground. The other six continue flying south at the original velocity. After the one bird has hit the ground, the centre of mass of all seven birds

(A) continues south at the original speed, but is now located some distance behind the flying birds

(B) continues south, but at 6/7 the original velocity

(C) continues south, but at 1/7 the original velocity

(D) stops with the dead bird **CM0039**

23. A non–zero external force acts on a system of particles. The velocity and acceleration of the centre of mass are found to be v_0 and a_C respectively at any instant t. It is possible that

(i) $v_0=0, a_C=0$ (ii) $v_0 \neq 0, a_C=0$ (iii) $v_0= 0, a_C\neq 0$ (iv) $v_0\neq 0, a_C\neq 0$

Then

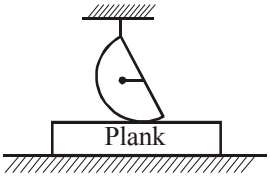
(A) (iii) and (iv) are true. (B) (i) and (ii) are true.

(C) (i) and (iii) are true. (D) (ii), (iii) and (iv) are true.

CM0042

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24. Lower surface of a plank is rough and lying at rest on a rough horizontal surface. Upper surface of the plank is smooth and has a smooth hemisphere placed over it through a light string as shown in the figure. After the string is burnt, trajectory of centre of mass of the sphere is :-



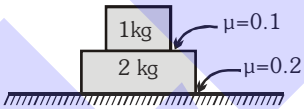
(A) a circle (B) an ellipse (C) a straight line (D) a parabola

CM0043

25. Two masses 'm_1' and 'm_2' ($m_1 > m_2$) are connected to the ends of a light inextensible string which passes over the surface of a smooth fixed pulley. If the system is released from rest, the acceleration of the centre of mass of the system will be (g = acceleration due to gravity) **CM0170**

(A) $\frac{g(m_1 - m_2)}{(m_1 + m_2)}$ (B) $\frac{g(m_1 - m_2)^2}{(m_1 + m_2)^2}$ (C) $\frac{g(m_1 + m_2)}{(m_1 - m_2)}$ (D) $\frac{g(m_1 + m_2)}{(m_1 - m_2)}$

26. If both the blocks as shown in the given arrangement are given together a horizontal velocity towards right. If a_{cm} be the subsequent acceleration of the centre of mass of the system of blocks then a_{cm} equals **CM0171**



(A) 0 m/s^2 (B) $\frac{5}{3}$ m/s^2 (C) $\frac{7}{3}$ m/s^2 (D) 2 m/s^2

27. Two particles of equal masses have velocities $\vec{v} = 4\hat{i}$ and $\vec{v} = 4\hat{j}$. First particle has an acceleration $\vec{a}_1 = (5\hat{i} + 5\hat{j})\,ms^{-2}$ while the acceleration of the other particle is zero. The centre of mass of the two particles moves in a path of **CM0172**

(A) Straight line (B) Parabola (C) Circle (D) Ellipse

28. A shell in flight explodes into n equal fragments k of the fragments reach the ground earlier than the other fragments. The acceleration of their centre of mass subsequently will be **CM0173**

(A) g (B) (n – k)g (C) $\frac{(n-k)g}{k}$ (D) $\frac{(n-k)}{n}g$

29. A body of mass 'm' is dropped and another body of mass M is projected vertically up with speed 'u' simultaneously from the top of a tower of height H . If the body reaches the highest point before the dropped body reaches the ground, then maximum height raised by the centre of mass of the system from ground is **CM0174**

(A) $H + \frac{u^2}{2g}$ (B) $\frac{u^2}{2g}$ (C) $H + \frac{1}{2g}\left(\frac{Mu}{m+M}\right)^2$ (D) $H + \frac{1}{2g}\left(\frac{mu}{m+M}\right)^2$

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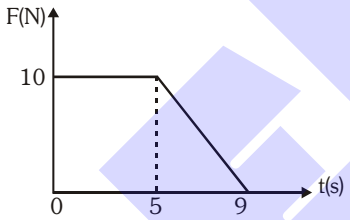
Momentum and law of conservation of momentum

30. A system of N particles is free from any external forces. **CM0175**
Which of the following is true for the magnitude of the total momentum of the system?
(A) It must be zero
(B) It could be non–zero, but it must be constant
(C) It could be non–zero, and it might not be constant
(D) The answer depends on the nature of the internal forces in the system

31. Which of the following must be true for the sum of the magnitudes of the momenta of the individual particles in the system?
(A) It must be zero
(B) It could be non–zero, but it must be constant
(C) It could be non–zero, and it might not be constant
(D) It could be zero, even if the magnitude of the total momentum is not zero

CM0045

32. A body of mass 4 kg is acted on by a force which varies as shown in the graph below. The momentum acquired is



(A) 280 N–s (B) 140 N–s (C) 70 N–s (D) 210 N–s

CM0046

33. Three interacting particles of masses 100 g, 200 g and 400 g each have a velocity of 20 m/s magnitude along the positive direction of x-axis, y-axis and z-axis. Due to force of interaction the third particle stops moving. The velocity of the second particle is $\left(10\hat{j}+5\hat{k}\right)$. What is the velocity of the first particle?

(A) $20\hat{i}+20\hat{j}+70\hat{k}$ (B) $10\hat{i}+20\hat{j}+8\hat{k}$ (C) $30\hat{i}+10\hat{j}+7\hat{k}$ (D) $15\hat{i}+5\hat{j}+60\hat{k}$

CM0044

34. Figure shows a block A of mass 5 kg kept at rest on a horizontal smooth surface. A spring (K = 200 N/m) which is compressed by 10 cm and tied with the help of a string to maintain the compression is attached to block A as shown in figure. Block B also of mass 5 kg moving with 2 m/s collides with A, as shown. During the collision the string breaks and after the collision the spring is in its natural state. Assume the bodies to be elastic and let the velocities of A and B be v_1 and v_2 respectively assuming positive direction towards right, after collision. Then

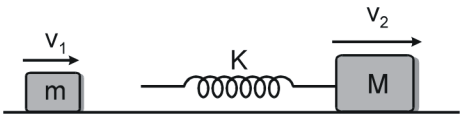


(A) $v_1 + v_2 > 2$
(B) Initial kinetic energy of system = final kinetic energy of system
(C) $v_1^2 + v_2^2 = 4.4\ (m/s)^2$
(D) $v_1 - v_2 = 2$

CM0091

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35. Two blocks of masses m and M are moving with speeds v_1 and v_2 ($v_1 > v_2$) in the same direction on the frictionless surface respectively, M being ahead of m. An ideal spring of force constant k is attached to the backside of M (as shown). The maximum compression of the spring when the block collides is: **CM0176**



(A) $v_1\sqrt{\frac{m}{k}}$ (B) $v_2\sqrt{\frac{M}{k}}$

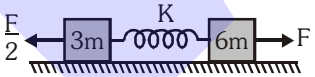
(C) $(v_1 - v_2)\sqrt{\frac{mM}{(M+m)\ K}}$ (D) None of above is correct.

36. Two blocks of mass 3 kg and 6 kg respectively are placed on a smooth horizontal surface. They are connected by a light spring of force constant k = 200 N/m. Initially the spring is unstretched. The indicated velocities are imparted to the blocks. Find the maximum extension of the spring.



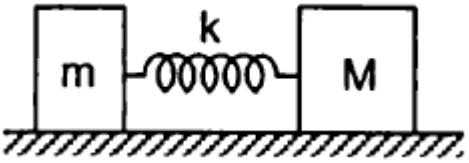
(A) 30 cm (B) 10 cm (C) $15\sqrt{2}$cm (D) 15 cm **CM0177**

37. For shown situation find the maximum elongation in the spring. Neglect friction everywhere. Initially, the blocks are at rest and spring is unstretched. **CM0178**



(A) $\frac{4F}{3K}$ (B) $\frac{3F}{4K}$ (C) $\frac{4F}{K}$ (D) $\frac{2F}{K}$

38. A light spring of spring constant k is kept compressed between two blocks of masses m and M on a smooth horizontal surface (figure). When released, the blocks acquire velocities in opposite directions. The spring loses contact with the blocks when it acquires natural length. If the spring was initially compressed through a distance x,find the final speeds of the two blocks. **CM0179**



(A) $V = \sqrt{\frac{km}{M(M+m)}}x$ (B) $V = \sqrt{\frac{k}{m+M}}x$

(C) $V = \sqrt{\frac{kM}{m(M+m)}}x$ (D) None of these

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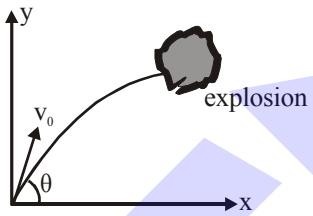
39. A particle of mass 4m which is at rest explodes into masses m, m & 2m. Two of the fragments of masses m and 2m are found to move with equal speeds v each in opposite directions. The total mechanical energy released in the process of explosion is **CM0180**

(A) mv^2 (B) $2mv^2$ (C) $1/2\ mv^2$ (D) $4mv^2$

40. Three identical particles moving with velocities $v_0\hat{i}, -3v_0\hat{j}$ and $5v_0\hat{k}$ collide successively with each other in such a way that they form a single particle. The velocity of resultant particle in i, j, k form is **CM0181**

(A) $v_0\left(\hat{i}-3\hat{j}+5\hat{k}\right)$ (B) $\frac{v_0}{3}\left(\hat{i}-3\hat{j}+5\hat{k}\right)$ (C) $\frac{v_0}{2}\left(\hat{i}-3\hat{j}+5\hat{k}\right)$ (D) $\frac{v_0}{3}\left(\hat{i}+3\hat{j}+5\hat{k}\right)$

41. A projectile is projected in x-y plane with velocity v_0. At top most point of its trajectory projectile explodes into two identical fragments. Both the fragments land simultaneously on ground and stick there. Taking point of projection as origin and R as range of projectile if explosion had not taken place. Which of the following can not be position vectors of two pieces, when they land on ground.



(A) $\frac{R}{2}\hat{i}, \frac{3R}{2}\hat{i}$ (B) $0\hat{i}, 2R\hat{i}$ (C) $R\hat{i}-R\hat{k}, R\hat{i}+R\hat{k}$ (D) $2R\hat{i}+\frac{R}{2}\hat{k}, R\hat{i}-\frac{R}{2}\hat{k}$

CM0048

Impulse

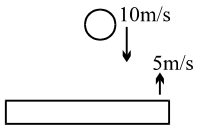
42. Velocity of a particle of mass 2 kg varies with time t according to the equation $\vec{v} = (2t\hat{i}+4\hat{j})\,m/s$. Here t is in seconds. Find the impulse imparted to the particle in the time interval from t = 0 to t = 2 s.

(A) $8\hat{i}\ N-s$ (B) $10\hat{i}\ N-s$ (C) $12\hat{i}\ N-s$ (D) $16\hat{i}\ N-s$ **CM0182**

43. A ball of mass 1 kg drops vertically on to the floor with a speed of 25 m/s. It rebounds with an initial velocity of 10 m/s. What impulse acts on the ball during contact? **CM0183**

(A) 35kg m/s downwards (B) 35 kg m/s upwards
(C) 30 kg m/s downwards (D) 30kg m/s upwards

44. A ball of mass 1kg strikes a heavy platform, elastically, moving upwards with a velocity of 5m/s. The speed of the ball just before the collision is 10m/s downwards. Then the impulse imparted by the platform on the ball is :-



(A) 15 N – s (B) 10 N – s (C) 20 N – s (D) 30 N – s

CM0058

45. Ball A of mass 5.0 kilograms moving at 20 m/s collides with ball B of unknown mass moving at 10m/s in the same direction. After the collision, ball A moves at 10 m/s and ball B at 15 m/s, both still in the same direction. What is the mass of ball B?

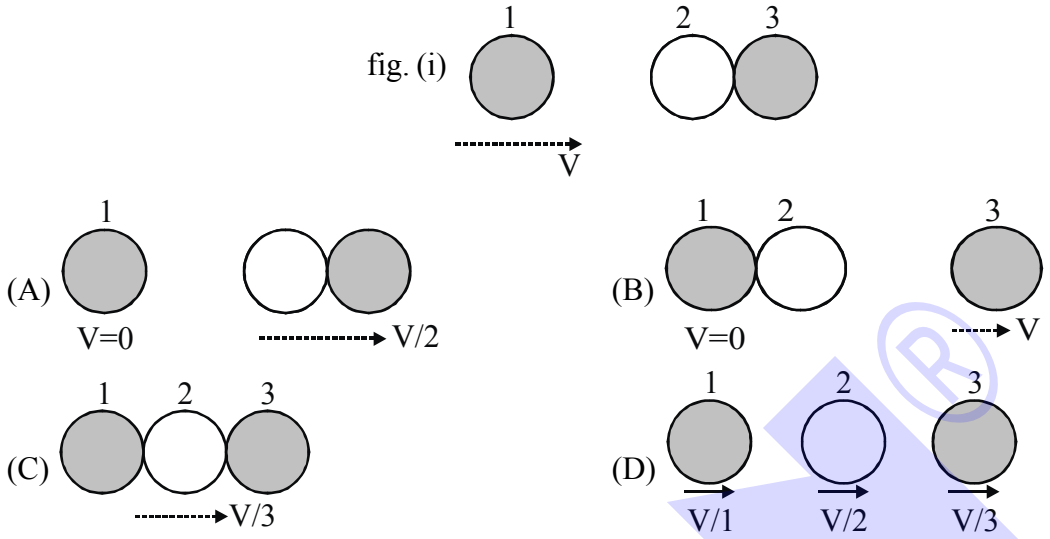
(A) 6.0 kg (B) 10. kg (C) 2.0 kg (D) 12 kg

CM0051

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Collision

46. Two identical ball bearings in contact with each other and resting on a frictionless table are hit head-on by another ball bearing of the same mass moving initially with a speed V as shown in figure (i). If the collision is elastic, which of the following is a possible result after collision?



CM0054

47. A ball is thrown downwards with initial speed = 6 m/s, from a point at height = 3.2 m above a horizontalf loor. If the ball rebounds back to the same height then coefficient of restitution equals to

(A) 1/2 (B) 0.75 (C) 0.8 (D) None

CM0056

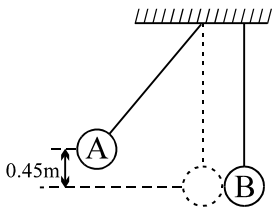
48. Two billiard balls undergo a head-on collision. Ball 1 is twice as heavy as ball 2. Initially, ball 1 moves with a speed v towards ball 2 which is at rest. Immediately after the collision, ball 1 travels at a speed of v/3 in the same direction. What type of collision has occured?

(A) inelastic (B) elastic

(C) completely inelastic (D) cannot be determined from the information given

CM0085

49. Two massless string of length 5 m hang from the ceiling very near to each other as shown in the figure. Two balls A and B of masses 0.25 kg and 0.5 kg are attached to the string. The ball A is released from rest at a height 0.45 m as shown in the figure. The collision between two balls is completely elastic. Immediately after the collision, the kinetic energy of ball B is 1 J. The velocity of ball A just after the collision is



(A) 5 ms^{-1} to the right (B) 5 ms^{-1} to the left (C) 1 ms^{-1} to the right (D) 1 ms^{-1} to the left

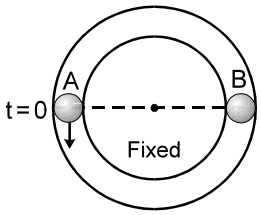
CM0087

50. A smooth small spherical ball of mass m, moving with velocity u collides head on with another small spherical ball of mass 3m, which was initially at rest. Two-third of the initial kinetic energy of the system is lost. The coefficient of restitution between the spheres is

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{2}$ (D) zero **CM0052**

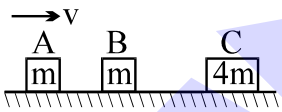
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51. Particle 'A' moves with speed 10 m/s in a frictionless circular fixed horizontal pipe of radius 5 m and strikes with 'B' of double mass that of A. Coefficient of restitution is 1/2 and particle 'A' starts its journey at t = 0. The time at which second collision occurs is : **CM0184**



(A) $\frac{\pi}{2}$ s (B) $\frac{2\pi}{3}$ s (C) $\frac{5\pi}{2}$ s (D) 4π s

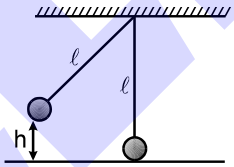
52. Three blocks are initially placed as shown in the figure. Block A has mass m and initial velocity v to the right. Block B with mass m and block C with mass 4m are both initially at rest. Neglect friction. All collisions are elastic. The final velocity of block A is



(A) 0.6v to the left (B) 1.4v to the left (C) v to the left (D) 0.4v to the right

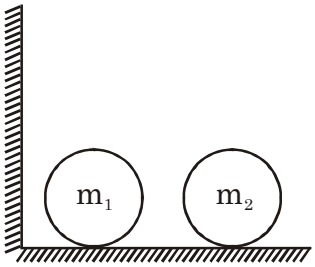
CM0084

53. In the arrangement shown, the pendulum on the left is pulled aside. It is then released and allowed to collide with other pendulum which is at rest. A perfectly inelastic collision occurs and the system rises to a height 1/4 h. The ratio of the masses of the pendulum is : **CM0185**



(A) 1 (B) 2 (C) 3 (D) 4

54. Mass m_1 strikes m_2 which is at rest. The ratio of masses m_1/m_2 for which they will collide again. (Collisions between ball and wall are elastic. Coefficient of restitution between m_1 and m_2 is e and all the surfaces are smooth) **CM0186**



(A) $\frac{3e}{2+e}$ (B) $\frac{2e}{2+e}$ (C) $\frac{e}{2(2+e)}$ (D) 1

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55. A 6 kg mass travelling at 2.5 ms^{-1} collides head on with a stationary 4 kg mass. After the collision the 6 kg mass travels in its original direction with a speed of 1ms^{-1}. The final velocity of 4 kg mass is:
(A) 1 ms^{-1} (B) 2.25 ms^{-1} (C) 2 ms^{-1} (D) 0 ms^{-1} **CM0187**

56. A block of mass 1 kg moving with a speed of 4 ms^{-1}, collides with another block of mass 2 kg which is at rest. The lighter block comes to rest after collision. The loss in KE of the system is: **CM0188**
(A) 8 J (B) 4×10^{-7} J (C) 4 J (D) 0 J

57. A ball is dropped from height 'H' onto a horizontal surface. If the coefficient of restitution is 'e' then the total time after which it comes to rest is: **CM0189**

(A) $\sqrt{\frac{2H}{g}}\left(\frac{1-e}{1+e}\right)$ (B) $\sqrt{\frac{2H}{g}}\left(\frac{1+e}{1-e}\right)$ (C) $\sqrt{\frac{2H}{g}}\left(\frac{1+e^2}{1-e^2}\right)$ (D) $\sqrt{\frac{2H}{g}}\left(\frac{1-e^2}{1+e^2}\right)$

58. A block of wood of mass 9.8 kg is suspended by a string. A bullet of mass 200 gm strikes horizontally with a velocity of 100ms^{-1} and gets embedded in it. The maximum height attained by the block is ($g = 10\ ms^{-2}$) **CM0190**
(A) 0.1 m (B) 0.2 m (C) 0.3 m (D) 0 m

59. A boy hits a baseball with a bat and imparts an impulse J to the ball. The boy hits the ball again with the same force, except that the ball and the bat are in contact for twice the amount of time as in the first hit. The new impulse equals:
(A) half the original impulse (B) the original impulse
(C) twice the original impulse (D) four times the original impulse **CM0049**

60. A ball of mass m approaches a moving wall of infinite mass with a speed 'v' along the normal to the wall. The speed of the wall is 'u' toward the ball. The speed of the ball after 'elastic' collision with wall is: **CM0191**
(A) u + v away from the wall (B) 2u + v away from the wall
(C) $|u - v|$ away from the wall (D) $|v - 2u|$ away from the wall

61. A ball collides with an inclined plane of inclination θ after falling through a distance h. If it moves horizontally just after the impact, the coefficient of restitution is : **CM0192**
(A) $\tan^2\theta$ (B) $\cot^2\theta$ (C) $\tan\theta$ (D) $\cot\theta$

62. A ball strikes a smooth horizontal ground at an angle of 45° with the vertical. What cannot be the possible angle of its velocity with the vertical after the collision. (Assume $e \le 1$).
(A) 45° (B) 30° (C) 53° (D) 60°
CM0053

63. A ball is projected from ground with a velocity V at an angle θ to the vertical. On its path it makes an elastic collison with a vertical wall and returns to ground. The total time of flight of the ball is

(A) $\frac{2v\sin\theta}{g}$ (B) $\frac{2v\cos\theta}{g}$ (C) $\frac{v\sin 2\theta}{g}$ (D) $\frac{v\cos\theta}{g}$

CM0055

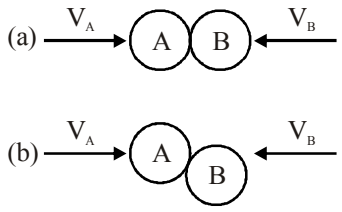
64. A particle is projected from a smooth horizontal surface with velocity v at an angle θ from horizontal. Coefficient of restitution between the surface and ball is e. The distance of the point where ball strikes the surface second time from the point of projection is

(A) $\frac{v^2\sin 2\theta(1+e^2)}{g}$ (B) $\frac{v^2\sin 2\theta(1+e^4)}{g}$ (C) $\frac{v^2\sin 2\theta(1+e^3)}{g}$ (D) $\frac{v^2\sin 2\theta(1+e)}{g}$

CM0057

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65. Two bodies, A and B, collide as shown in figures a and b below. Circle the true statement :



(A) They exert equal and opposite forces on each other in (a) but not in (b)
(B) They exert equal and opposite force on each other in (b) but not in (a)
(C) They exert equal and opposite force on each other in both (a) and (b)
(D) The forces are equal and opposite to each other in (a), but only the components of the forces parallel to the velocities are equal in (b).

CM0059

66. A mass 'm' moves with a velocity 'v' and collides inelastically with another identical mass at rest. After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision :-

(A) $\frac{2v}{\sqrt{3}}$

(B) $\frac{v}{\sqrt{3}}$

(C) $v\sqrt{\frac{2}{3}}$

(D) the situation of the problem is not possible without external impulse

CM0060

67. A ball of mass m moving at a speed v makes a head-on collision with an identical ball at rest. The kinetic energy of the balls after the collision is three fourths of the original. Find the coefficient of restitution. **CM0193**

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\sqrt{\frac{2}{3}}$

Variable mass and collision of a continuous stream

68. A bead of mass m and diameter d is sliding back and forth with velocity v on a wire held between two rigid walls of length L. Assume that the collisions with the wall are perfectly elastic and there is no friction. The average force that the bouncing bead exerts on the one of the walls is **CM0194**

(A) $\frac{mv^2}{L-d}$ (B) $\frac{mv^2}{L+d}$ (C) $\frac{2mv^2}{L-d}$ (D) $\frac{2mv^2}{L+d}$

69. A continuous stream of particles of mass m and velocity v, is emitted from a source at a rate of n per second. The particles travel along a straight line, collide with a body of mass M and get embedded in the body. If the mass M was originally at rest, its velocity when it has received N particles will be

(A) $\frac{mvn}{Nm+n}$ (B) $\frac{mvN}{Nm+M}$ (C) $\frac{mv}{Nm+M}$ (D) $\frac{Nm+M}{mv}$ **CM0195**

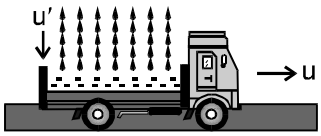
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70. An open water tight railway wagon of mass 5×10^3 kg coasts at an initial velocity 1.2 m/s without friction on a railway track. Rain drops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be

(A) 0.5 m/s (B) 2 m/s (C) 1 m/s (D) 1.5 m/s

CM0092

71. The carriage of mass M has constant initial velocity u along a straight horizontal track when at t = 0, it starts raining. The rain drops have a vertical velocity u′ and result into addition of mass m per second to the carriage. The velocity of car after T second of start of rain is (Assume frictionless surface) : **CM0196**



(A) $\frac{Mu}{M+mt}$ (B) $\frac{Mu+mu'}{M+mt}$ (C) $\frac{(u+u')(M+m)}{M}$ (D) $\frac{M(u+U')}{mt}$

72. A wagon filled with sand has a hole so that sand leaks through the bottom at a constant rate λ. An external force $\vec{F}$ acts on the wagon in the direction of motion. Assuming instantaneous velocity of the wagon to be $\vec{v}$ and initial mass of system to be m_0, the force equation governing the motion of the wagon is : **CM0197**

(A) $\vec{F} = m_0 \frac{d\vec{v}}{dt} + \lambda \vec{v}$ (B) $\vec{F} = m_0 \frac{d\vec{v}}{dt} - \lambda \vec{v}$

(C) $\vec{F} = (m_0 - \lambda t) \frac{d\vec{v}}{dt}$ (D) $\vec{F} = (m_0 - \lambda t) \frac{d\vec{v}}{dt} + \lambda \vec{v}$

73. A balloon having mass ' m ' is filled with gas and is held in hands of a boy. Then suddenly it get released and gas starts coming out of it with a constant rate. The velocities of the ejected gases is also constant 2 m/s with respect to the balloon. Find out the velocity of the balloon when the mass of gas is reduced to half. (neglect gravity) **CM0198**

(A) ℓ n 2 (B) 2 ℓn 4 (C) 2 ℓn 2 (D) none of these

74. A rocket of mass 4000 kg is set for vertical firing. How much gas must be ejected per second so that the rocket may have initial upwards acceleration of magnitude 19.6 m/s^2. [Exhaust speed of fuel = 980 m/s.]

(A) 240 kg s^{-1} (B) 60 kg s^{-1} (C) 120 kg s^{-1} (D) None

CM0093

MULTIPLE CORRECT TYPE QUESTIONS

Center of mass

75. The centre of mass of a system of particles is at the origin. Which of the following statements is incorrect. **CM0199**

(A) The number of particles to the right of the origin is equal to the number of particles to the left

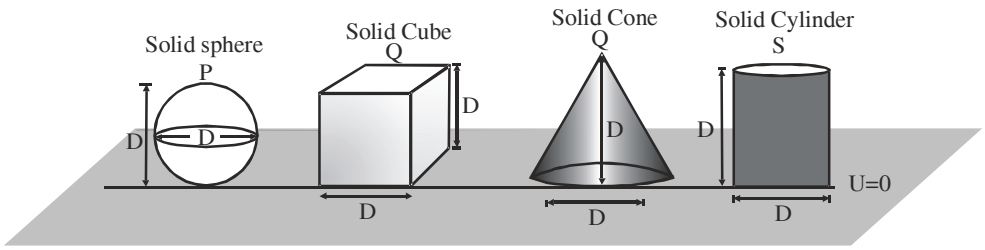
(B) The total mass of the particles to the right of the origin is same as the total mass to the left of the origin

(C) The number of particles on X-axis should be equal to the number of particles on Y-axis

(D) If there is a particle on the positive X-axis, there must be at least one particle on the negative X-axis.

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76. Assuming potential energy 'U' at ground level to be zero.



All objects are made up of same material.

U_P = Potential energy of solid sphere U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone U_S = Potential energy of solid cylinder

(A) $U_S > U_P$ (B) $U_Q > U_S$ (C) $U_P > U_Q$ (D) $U_S > U_R$

CM0094

77. A man of mass 80 kg stands on a plank of mass 40 kg. The plank is lying on a smooth horizontal floor. Initially both are at rest. The man starts walking on the plank towards north and stops after moving a distance of 6 m on the plank. Then **CM0200**

(A) The centre of mass of plank-man system remains stationary.

(B) The plank will slide to the north by a distance 4 m

(C) The plank will slide to the south by a distance 4 m

(D) The plank will slide to the south by a distance 12 m

78. For a two-body system in absence of external forces, the kinetic energy as measured from ground frame is K_0 and from centre mass frame is K_{cm}. Choose the **CORRECT** statement(s) :- **CM0201**

(A) The kinetic energy as measured from center of mass frame is least

(B) Only the portion of energy K_{cm} can be transformed from one form to another due to internal changes in the system.

(C) The system always retains at least $K_0 - K_{cm}$ amount of kinetic energy as measured from ground frame irrespective of any kind of internal changes in the system.

(D) The system always retains at least K_{cm} amount of kinetic energy as measured from ground frame irrespective of any kind of internal changes in the system.

CM frame of reference

79. In head on elastic collision of two bodies of equal masses **CM0202**

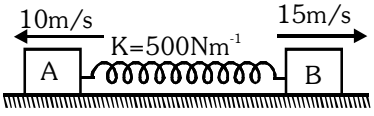
(A) the velocities are interchanged

(B) the speeds are interchanged

(C) the momenta are interchanged

(D) the faster body slows down and the slower body speeds up.

80. Two blocks A and B are joined together with a compressed spring. When the system is released, the two blocks appear to be moving with unequal speeds in the opposite directions as shown in figure. Select incorrect statement(s) :



(A) The centre of mass of the system will remain stationary.

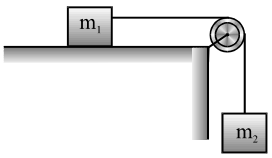
(B) Mass of block A is equal to mass of block B.

(C) The centre of mass of the system will move towards right.

(D) It is an impossible physical situation. **CM0203**

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81. X and Y components of acceleration of C. M. are :



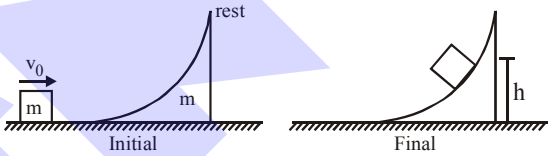
(A) $(a_{CM})_X = \frac{m_1 m_2 g}{m_1 + m_2}$ (B) $(a_{CM})_X = \frac{m_1 m_2 g}{(m_1 + m_2)^2}$

(C) $(a_{CM})_Y = \left(\frac{m_2}{m_1 + m_2}\right)^2 g$ (D) $(a_{CM})_Y = \left(\frac{m_2}{m_1 + m_2}\right) g$ **CM0204**

Momentum conservation

82. Two charges moving under their only own mutual attraction separated by large distance initially. Then choose the correct statement(s)
(A) If both are free, mechanical energy is conserved.
(B) If one is fixed and other is free, mechanical energy is conserved.
(C) If one is fixed and other is free, momentum is conserved.
(D) If both are free momentum is conserved. **CM0061**

83. In the arrangement shown, horizontal surface is smooth, but friction is present between the block and the surface of the wedge. Block is given velocity v_0 at $t = 0$. After achieving height 'h' on the wedge, block comes to rest with respect to wedge at $t = t_0$. Then from $t = 0$ to $t = t_0$:-
(A) Work done by friction on the block is negative
(B) Work done by friction on the wedge is negative
(C) Work done by block on the wedge is positive
(D) Work done by wedge on the block is positive



CM0062

84. A blast breaks a body initially at rest of mass 0.5 kg into three pieces, two smaller pieces of equal mass and the third double the mass of either of small piece. After the blast the two smaller masses move at right angles to one another with equal speed. Find the statements that is/are true for this case assuming that the energy of blast is totally transferred to masses.
(A) All the three pieces share the energy of blast equally
(B) The speed of bigger mass is $\sqrt{2}$ times the speed of either of the smaller mass
(C) The direction of motion of bigger mass makes an angle of 135° with the direction of smaller pieces
(D) The bigger piece carries double the energy of either piece. **CM0095**

Collisions

85. A particle moving with kinetic energy = 3 joule makes an elastic head on collision with a stationary particle which has twice its mass during the impact.
(A) The minimum kinetic energy of the system is 1 joule.
(B) The maximum elastic potential energy of the system is 2 joule.
(C) Momentum and total kinetic energy of the system are conserved at every instant.
(D) The ratio of kinetic energy to potential energy of the system first decreases and then increases.
CM0096

86. A ball moving with a velocity v hits a massive wall moving towards the ball with a velocity u. An elastic impact lasts for a time Δt. **CM0205**

(A) The average elastic force acting on the ball is $\frac{m(u+v)}{\Delta t}$

(B) The average elastic force acting on the ball is $\frac{2m(u+v)}{\Delta t}$

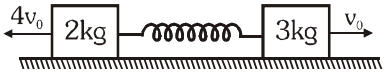
(C) The kinetic energy of the ball increases by $2mu(u+v)$
(D) The kinetic energy of the ball remains the same after the collision

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COMPREHENSION TYPE QUESTIONS

Paragraph for Question 87 to 89

2 kg and 3 kg blocks are placed on a smooth horizontal surface and connected by spring which is unstretched initially. The blocks are imparted velocities as shown in the figure.



87. The maximum energy stored in the spring in the subsequent motion will be

(A) $5v_0^2$ (B) $15v_0^2$ (C) zero (D) $10v_0^2$

CM0066

88. Maximum speed of 3 kg block in the subsequent motion will be

(A) v_0 (B) $2v_0$ (C) $3v_0$ (D) $4v_0$

CM0066

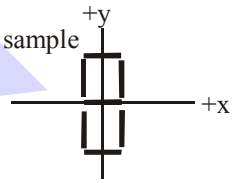
89. Maximum speed of 2 kg block in the subsequent motion will be

(A) v_0 (B) $2v_0$ (C) $3v_0$ (D) $4v_0$

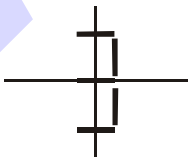
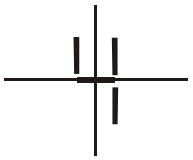
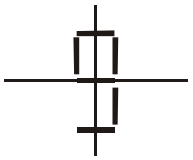
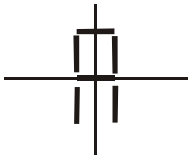
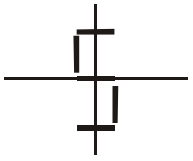
CM0066

MATRIX MATCH TYPE QUESTIONS

90. On the left are statements about the location of the center of mass of the objects depicted on the right. The objects on the right are symbols constructed out of sticks of equal length and mass. The location of the center of mass is described using the coordinate system depicted in the sample.



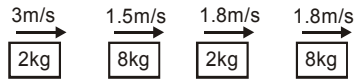
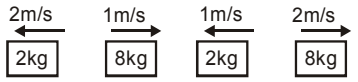
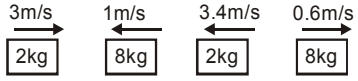
The centre of mass lies at x = 0, y = 0

Column I	Column II
(A) The center of mass is at x > 0 and y = 0	(P) 
(B) The center of mass is at x = 0 and y > 0	(Q) 
(C) The center of mass is at x > 0 and y > 0	(R) 
(D) The center of mass is at x = 0 and y = 0	(S) 
	(T) 

CM0067

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91. In Column-I, 4 situations are depicted and in column-II, 4 possible kinds of collision are listed. Match the situation with type of collision.

Column-I **Before**		**Column-II** **After**	
(A)		(P)	Elastic
(B)		(Q)	Perfectly Inelastic
(C)		(R)	Partially elastic
		(S)	Collision is not possible

CM0069

92. A particle of mass m, kinetic energy K and momentum p collides head on elastically with another particle of mass 2 m at rest. After collision :

	Column I		**Column II**
(A)	Momentum of first particle	(P)	3/4 p
(B)	Momentum of second particle	(Q)	– K/9
(C)	Kinetic energy of first particle	(R)	– p/3
(D)	Kinetic energy of second particle	(S)	$\frac{8K}{9}$
		(T)	None

CM0070

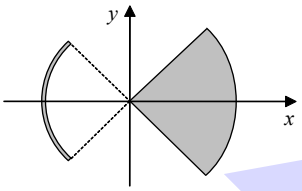
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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

Centre of mass

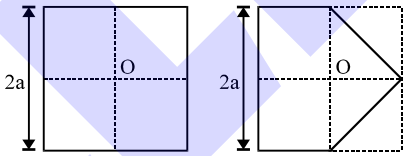
1. A sector cut from a uniform disk of radius 12 cm and a uniform rod of the same mass bent into shape of an arc are arranged facing each other as shown in the figure. If center of mass of the combination is at the origin, what is the radius of the arc?



(A) 8 cm (B) 9 cm (C) 12 cm (D) 18 cm

CM0071

2. A piece of paper (shown in figure-1) is in form of a square. Two corners of this square are folded to make it appear like figure-2. Both corners are put together at centre of square 'O'. If O is taken to be (0, 0), the centre of mass of new system will be at

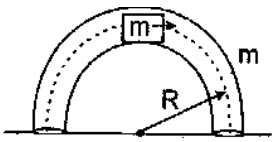


(A) $\left(\frac{-a}{8}, 0\right)$ (B) $\left(\frac{-a}{6}, 0\right)$ (C) $\left(\frac{a}{12}, 0\right)$ (D) $\left(\frac{-a}{12}, 0\right)$

CM0072

Motion f CM & CM frame of reference

3. In a vertical plane inside a smooth hollow thin tube a block of same mass as that of tube is released as shown in figure. When it is slightly disturbed, it moves towards right. By the time the block reaches the right end of the tube then the displacement of the tube will be (where 'R' is mean radius of tube). Assume that the tube remains in vertical plane. **CM0206**



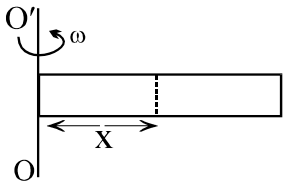
(A) $\frac{2R}{\pi}$ (B) $\frac{4R}{\pi}$ (C) $\frac{R}{2}$ (D) R

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll\8.Rot Dy\Hindi\03_COM & Collision\01_Hindi.p65

4. A particle of mass 2m is connected by an inextensible string of length 1.2 m to a ring of mass m which is free to slide on a horizontal smooth rod. Initially the ring and the particle are at the same level with the string taut. Both are then released simultaneously. The distance in meters moved by the ring when the string becomes vertical is **CM0207**

(A) 0 (B) 0.4 (C) 0.8 (D) 1.2

5. A uniform rod of length l and mass m is allowed to rotate in a horizontal plane about a vertical axis OO′ passing through one of its ends with an angular velocity ω. Then the tension in the rod in terms of distance x measured from the axis of rotation is: **CM0208**



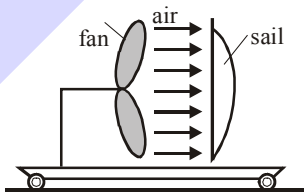
(A) $\frac{1}{2}m\omega^2 l\left(1-\frac{x^2}{l^2}\right)$ (B) $m\omega^2 l\left(1-\frac{x^2}{l^2}\right)$ (C) $\frac{1}{2}m\omega^2 l\left(1+\frac{x^2}{l^2}\right)$ (D) $m\omega^2 l\left(1+\frac{x^2}{l^2}\right)$

Momentum conservation

6. A test tube of mass 20 gm is filled with a gas and fitted with a stopper of 2gm. It is suspended horizontally by means of a thread of 1m length and heated. When the stopper kicks out, the tube just completes a circle in vertical plane. The velocity with which the stopper kicked out is **CM0209**

(A) $7ms^{-1}$ (B) $10\ ms^{-1}$ (C) $70\ ms^{-1}$ (D) $0.1ms^{-1}$

7. A fan and a sail are mounted vertically on a cart that is initially at rest on a horizontal table as shown in the diagram. When the fan is turned on, an air stream is blown towards the right and is incident on the sail. The cart is free to move with negligible resistance forces. After the fan has been turned on the cart will



(A) move to the right and then to the left (B) remain at rest
(C) move towards the right (D) move towards the left

CM0073

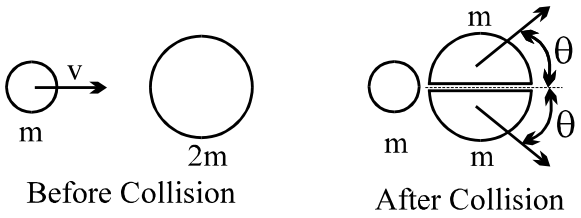
8. An isolated particle of mass m is moving in horizontal plane (x–y), along the x–axis, at a certain height above the ground. It suddenly explodes into two fragment of masses $\frac{m}{4}$ and $\frac{3m}{4}$. An instant later, the smaller fragment is at y = +15 cm. The larger fragment at this instant is at :–

(A) y = –5 cm (B) y = +20 cm (C) y = +5 cm (D) y = –20 cm

CM0079

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9. A particle of mass m is moving along the x-axis with speed v when it collides with a particle of mass 2m initially at rest. After the collisions, the first particle has come to rest, and the second particle has split into two equal-mass pieces that move at equal angles $\theta > 0$ with the x-axis, as shown in the figure. Which of the following statements correctly describes the speeds of the two pieces?



(A) Each piece moves with speed v

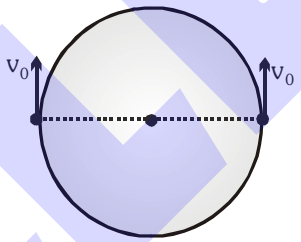
(B) One of the pieces moves with speed v, the other moves with speed less than v

(C) Each piece moves with speed $v/2$

(D) Each piece moves with speed greater than $v/2$

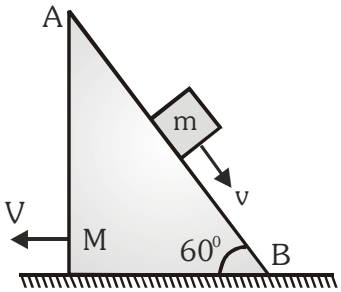
CM0080

10. Two particles of mass m, constrained to move along the circumference of a smooth circular hoop of equal mass m, are initially located at opposite ends of a diameter and given equal velocities v_0 shown in the figure. The entire arrangement is located in gravity free space. Their velocity just before collision is:



(A) $\frac{1}{\sqrt{3}}v_0$ (B) $\frac{\sqrt{3}}{2}v_0$ (C) $\frac{2}{\sqrt{3}}v_0$ (D) $\frac{\sqrt{7}}{3}v_0$ **CM0210**

11. A particle of mass m = 0.1 kg is released from rest from a point A of a wedge of mass M = 2.4 kg free to slide on a frictionless horizontal plane. The particle slides down the smooth face AB of the wedge. When the velocity of the wedge is 0.2 m/s the velocity of the particle in m/s relative to the wedge is



(A) 4.8 (B) 5 (C) 7.5 (D) 10 **CM0211**

ALLEN®

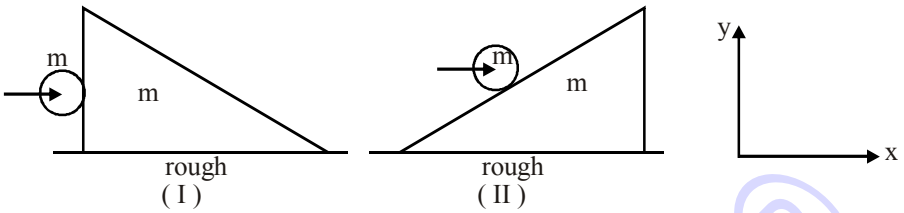
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Impulse

12. A ball of mass m collides horizontally with a stationary wedge on a rough horizontal surface, in the two orientations as shown. Neglect friction between ball and wedge. Two student comment on system of ball and wedge in these situations

Saurav : Momentum of system in x-direction will change by significant amount in both cases.

Rahul : There are no impulsive external forces in y-direction in both cases hence the total momentum of system in y-direction can be treated as conserved in both cases.



(A) **Saurav** is wrong and **Rahul** is correct (B) **Saurav** is correct and **Rahul** is wrong

(C) Both are correct (D) Both are wrong

CM0081

13. A force exerts an impulse I on a particle changing its speed from u to 2u. The applied force and the initial velocity are oppositely directed along the same line. The work done by the force is

(A) $\frac{3}{2}Iu$ (B) $\frac{1}{2}Iu$ (C) I u (D) 2 I u

CM0083

Collision

14. A smooth sphere is moving on a horizontal surface with a velocity vector $(2\hat{i}+2\hat{j})$ m/s immediately before it hit a vertical wall. The wall is parallel to vector $\hat{j}$ and coefficient of restitution between the sphere and the wall is $e = 1/2$. The velocity of the sphere after it hits the wall is

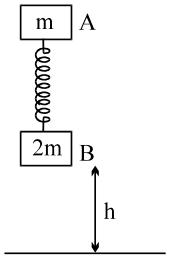
(A) $\hat{i}-\hat{j}$ (B) $-\hat{i}+2\hat{j}$ (C) $-\hat{i}-\hat{j}$ (D) $2\hat{i}-\hat{j}$

CM0089

15. A tennis ball bounces down a flight of stairs, striking each step in turn and rebounding to half of height of the step. The coefficient of restitution is **CM0212**

(A) 1/2 (B) $\frac{1}{\sqrt{2}}$ (C) $\left(\frac{1}{\sqrt{2}}\right)^{1-2}$ (D) $\left(\frac{1}{\sqrt{2}}\right)^{1-4}$

16. From what minimum height h must the system be released when spring is unstretched so that after perfectly inelastic collision (e = 0) with ground, B may be lifted off the ground (Spring constant = k).



(A) mg/(4k) (B) 4mg/k (C) mg/(2k) (D) none

CM0078

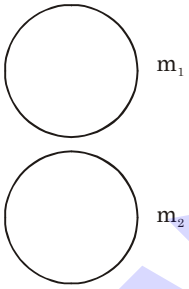
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17. Two balls of same mass are dropped from the same height h, on to the floor. The first ball bounces to a height h/4 ,after the collision & the second ball to a height h/16. The impulse applied by the first & second ball on the floor are I_1 and I_2 respectively. Then

(A) $5I_1 = 6I_2$ (B) $6I_1 = 5I_2$ (C) $I_1 = 2I_2$ (D) $2I_1 = I_2$

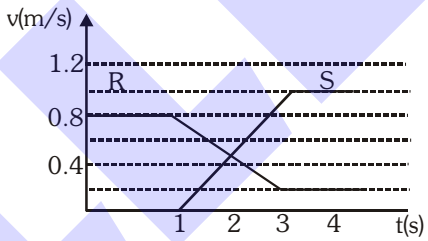
CM0050

18. Two balls of masses m_1 and m_2 are placed on top of one over the other (with a small gap between them) and then dropped on to the ground. What is the ratio m_1/m_2 for which the upper ball ultimately receives the largest possible fraction of the total energy? Take all collisions as elastic. Neglect air resistance.



(A) 1:1 (B) 1: 2 (C) 1:3 (D) 1:4 **CM0213**

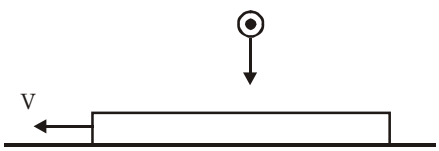
19. The diagram shows the velocity–time graph for two masses R and S that collided elastically. Which of the following statements is true ?



I. R and S moved in the same direction after the collision
II. The velocities of R and S were equal at the mid time of the collision.
III. The mass of in was greater than mass of S. **CM0214**

(A) I only (B) II only (C) I and II only (D) I, II and III

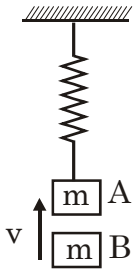
20. A board is moving with velocity v on a smoother horizontal plane. The upper surface of the board is rough on which a ball falls with velocity v and rebounds with velocity $\frac{v}{2}$. The mass of the board is same as that of ball. After the collision , the board comes to state of rest. The co-efficient of friction between the board and the ball can be **CM0215**



(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) $\frac{3}{5}$

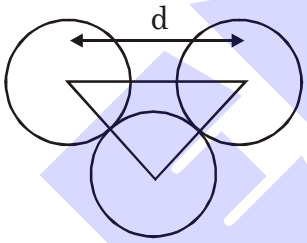
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21. Block A is hanging from a vertical spring of spring constant k and is at rest. Block B strikes block A with velocity v and sticks to it. Then the value of v for which the spring just attains natural length is **CM0216**



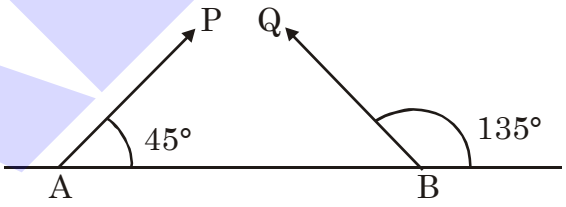
(A) $\sqrt{\frac{60mg^2}{k}}$ (B) $\sqrt{\frac{6mg^2}{k}}$ (C) $\sqrt{\frac{10mg^2}{k}}$ (D) $\sqrt{\frac{mg^2}{k}}$

22. Two identical ball of radii r are kept on a horizontal plane with their centers d distance apart. A third ball, identical to previous one, collide elastically with both the balls symmetrically as shown in the figure. If the third ball comes to rest after the collision, d should be **CM0217**



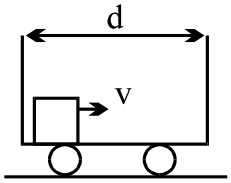
(A) 3r (B) $2\sqrt{2}r$ (C) $\left(\sqrt{2}+1\right)r$ (D) $\left(\sqrt{2}+2\right)r$

23. Particles P and Q of masses 20g and 40g, respectively, are projected from positions A and B on the ground. The initial velocities of P and Q make angles of 45^0 and 135^0, respectively with the horizontal as shown in the Fig. Each particle has an initial speed of 49m/s. The separation AB is 245m. Both particles travel in the same vertical plane and undergo a collision. After the collision P retraces its path. The separation of Q from its initial position when it hits the ground is **CM0218**



(A) 245 m (B) $\frac{245}{3}m$ (C) $\frac{245}{2}m$ (D) $\frac{245}{\sqrt{2}}m$

24. In a smooth stationary cart of length d, a small block is projected along it's length with velocity v towards front. Coefficient of restitution for each collision is e. The cart rests on a smooth ground and can move freely. The time taken by block to come to rest w.r.t. cart is



(A) $\frac{ed}{(1-e)v}$ (B) $\frac{ed}{(1+e)v}$

(C) $\frac{d}{e}$ (D) infinite

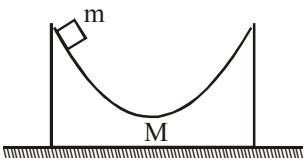
CM0088

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MULTIPLE CORRECT TYPE QUESTIONS

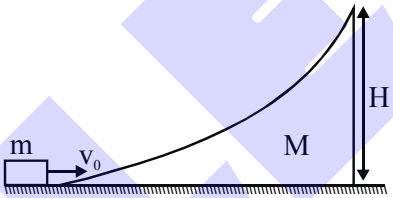
Momentum conservation

25. Figure shows a wedge on which a small block is released from rest. All the surfaces are smooth system comprises of wedge and blocks. Mark the correct statement(s) regarding motion of block on wedge till block attains maximum height on wedge.



(A) Acceleration of centre of mass of system is initially vertically down then vertically up.
(B) Initially centre of mass moves down and then up.
(C) At the maximum height block and wedge move with common velocity.
(D) Centre of mass of wedge moves towards left then right **CM0063**

26. Figure shows a block of mass m projected with velocity v_0 towards a wedge. Consider all the surfaces to be smooth. Block does not have sufficient energy to negotiate (over come) wedge. Mark the correct option(s)



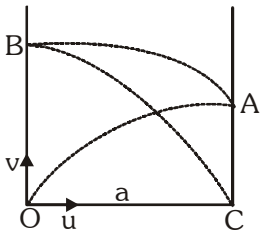
(A) when block is at the maximum height on wedge, block and wedge have velocity equal to velocity of centre of mass of block wedge system
(B) wedge acquires maximum speed with respect to ground when block returns to lowest point on wedge.
(C) momentum of wedge and block is conserved at all times
(D) centre of mass of wedge and block remains stationary. **CM0064**

Collision

27. In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.
(A) The total momentum of the 'A plus B' system is p before and after the impact, and (p-J) during the impact.
(B) During the impact A gives impulse of magnitude J to B
(C) The coefficient of restitution is $\frac{2J}{p}-1$
(D) The coefficient of restitution is $\frac{J}{p}+1$ **CM0097**

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28. A ball is projected from a point in one of the two smooth parallel vertical walls against the other in a plane perpendicular to both after being reflected at each wall impinge again on the second at a point in the same horizontal plane as is started. The distance between two walls is a,b is the free range on a horizontal plane and e be the coefficient of restitution



(A) The total time taken in moving from O to C is $\frac{a}{e^2u}(e^2 + e + 1)$

(B) The free range on the horizontal plane b = $\frac{2uv}{g}$

(C) $be^2 = a(e^2 + e + 1)$

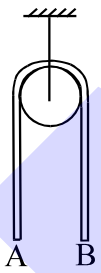
(D) All above options are correct

CM0219

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 29 and 30

A uniform chain of length 2L is hanging in equilibrium position, if end B is given a slightly downward displacement the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible



29. The acceleration of end B when it has been displaced by distance x, is

(A) $\frac{x}{L}g$ (B) $\frac{2x}{L}g$ (C) $\frac{x}{2}g$ (D) g

CM0099

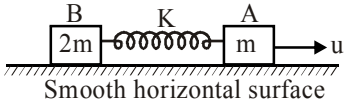
30. The velocity v of the string when it slips out of the pulley (height of pulley from floor > 2L)

(A) $\sqrt{\frac{gL}{2}}$ (B) $\sqrt{2gL}$ (C) $\sqrt{gL}$ (D) none of these

CM0099

MATRIX MATCH TYPE QUESTION

31. Two blocks A and B of mass m and 2m respectively are connected by a massless spring of spring constant K. This system lies over a smooth horizontal surface. At t = 0 the block A has velocity u towards right as shown while the speed of block B is zero, and the length of spring is equal to its natural length at that instant.

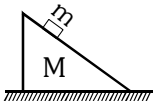
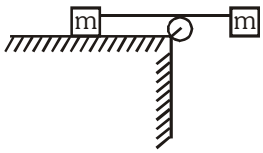
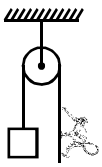
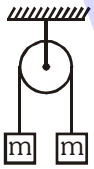


	Column-I		Column-II
(A)	The velocity of block A	(P)	can never be zero
(B)	The velocity of block B	(Q)	may be zero at certain instants of time
(C)	The kinetic energy of system of two block	(R)	is minimum at maximum compression of spring
(D)	The potential energy of spring	(S)	is maximum at maximum extension of spring

CM0101

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32. In each situation of column–I, a system involving two bodies is given. All strings and pulleys are light and friction is absent everywhere. Initially each body of every system is at rest. Consider the system in all situation of column I from rest till any collision occurs. Then match the statements in column – I with the corresponding results in column–II

Column I	**Column II**
(A) The block plus wedge system is placed over smooth horizontal surface. After the system is released from rest, the centre of mass of system 	(P) Shifts towards right
(B) The string connecting both the blocks of mass m is horizontal. Left block is placed over smooth horizontal table as shown. After the two block system is released from rest, the centre of mass of system 	(Q) Shifts downwards
(C) The block and monkey have same mass. The monkey starts climbing up the rope. After the monkey starts climbing up, the centre of mass of monkey+block system 	(R) Shifts upwards
(D) Both block of mass m are initially at rest. The left block is given initial velocity u downwards. Then, the centre of mass of two block system afterwards 	(S) Does not shift

CM0100

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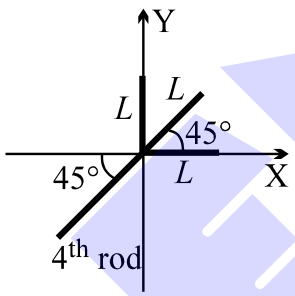
EXERCISE (S)

Center of mass

1. Four particles of mass 5, 3, 2, 4 kg are at the points (1, 6), (–1, 5), (2, –3), (–1, –4). Find the coordinates of their centre of mass.

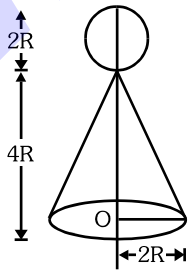
CM0001

2. Three identical uniform rods of the same mass *M* and length *L* are arranged in xy plane as shown in the figure. A fourth uniform rod of mass *3M* has been placed as shown in the xy plane. What should be the value of the length of the fourth rod such that the center of mass of all the four rods lie at the origin?



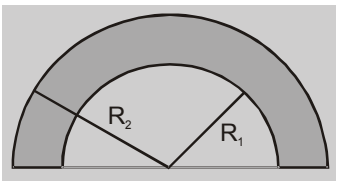
CM0003

3. A man has constructed a toy as shown in figure. If density of the material of the sphere is 12 times of the cone compute the position of the centre of mass. [Centre of mass of a cone of height h is at height of h/4 from its base.]



CM0005

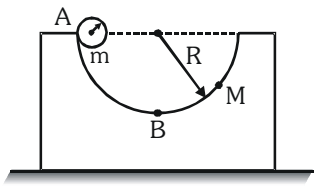
4. Find the centre of mass of a uniform plate having semicircular inner and outer boundaries of radii R_1 and R_2 (figure). **CM0220**



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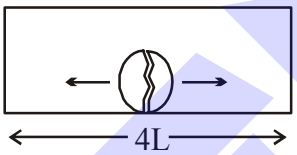
Motion of CM and CM frame of reference

5. A block of mass M with a semicircular track of radius R, rests on a horizontal frictionless surface. A uniform cylinder of radius r and mass m is released from rest at the top point A (see Fig). The cylinder slips on the semicircular frictionless track. How far has the block moved when the cylinder reaches the bottom (point B) of the track ? How fast is the block moving when the cylinder reaches the bottom of the track?



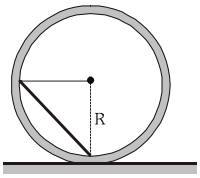
CM0027

6. A bomb of mass 3m is kept inside a closed box of mass 3m and length 4L at it's centre. It explodes in two parts of mass m & 2m. The two parts move in opposite direction and stick to the opposite side of the walls of box. Box is kept on a smooth horizontal surface. What is the distance moved by the box during this time interval.



CM0009

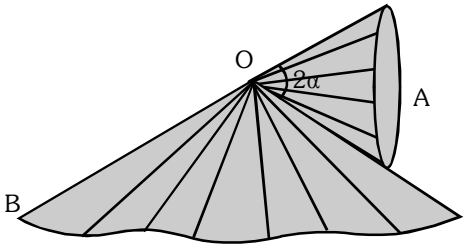
7. Inside a hollow uniform sphere of inner radius R a uniform rod of length $R\sqrt{2}$ is released from the state of rest as shown. The mass of the rod is same as that of the sphere. Assume friction to be absent everywhere. Horizontal displacement of sphere with respect to earth in the time in which the rod becomes horizontal, is



CM0026

8. Two small discs of masses m_1 and m_2 interconnected by a weightless spring rest on a smooth horizontal plane. The discs are set in motion with initial velocities v_1 and v_2 whose directions are mutually perpendicular and lie in a horizontal plane. Find the total energy $\tilde{E}$ of this system in the frame of the centre of mass. **CM0221**

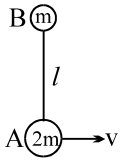
9. A round cone A of mass m = 3.2 kg and half angle $\alpha = 10°$ rolls uniformly and without slipping along a round conical surface B so that its apex O remains stationary. The centre of gravity of the cone A is at the same level as the point O and at a distance ℓ = 17 cm from it. The cone's axis moves with angular velocity ω. Find:



(a) the static friction force acting on the cone A, if ω = 1.0 rad/s;
(b) at what values of ω the cone A will roll without sliding, if the coefficient of friction between the surfaces is equal to k = 0.25. **CM0222**

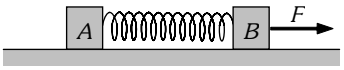
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10. Two masses A and B connected with an inextensible string of length ℓ lie on a smooth horizontal plane. A is given a velocity of v m/s along the ground perpendicular to line AB as shown in figure. Find the tension in string during their subsequent motion.



CM0029

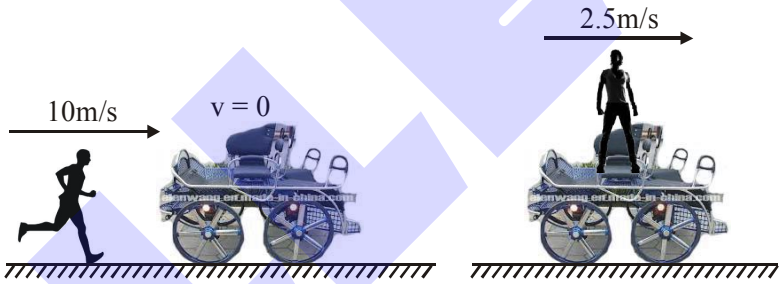
11. Two blocks each of mass m, connected by an un-stretched spring are kept at rest on a frictionless horizontal surface. A constant force F is applied on one of the blocks pulling it away from the other as shown in figure. **CM0223**



(a) Find acceleration of the mass center.
(b) Find the displacement of the centre of mass as function of time t.
(c) If the extension of the spring is x_0 at an instant t, find the displacements of the two blocks relative to the ground at this instant.

Momentum conservation

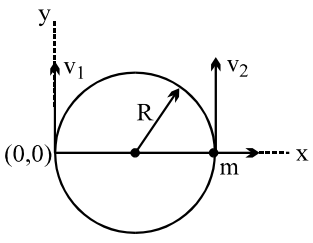
12. A 50 kg boy runs at a speed of 10 m/s and jumps onto a cart as shown in the figure. The cart is initially at rest. If the speed of the cart with the boy on it is 2.50 m/s, what is the mass of the cart ? (Assuming friction is absent between cart and ground)



CM0011

13. Two cars initially at rest are free to move in the x direction. Car A has mass 4 kg and car B has mass 2 kg. They are tied together, compressing a spring in between them. When the spring holding them together is burned, car A moves off with a speed of 2 m/s.
(i) With what speed does car B leave.
(ii) How much energy was stored in the spring before it was burned. **CM0012**

14. A particle of mass m, moving in a circular path of radius R with a constant speed v_2 is located at point (2R, 0) at time t = 0 and a man starts moving with a velocity v_1 along the +ve y-axis from origin at time t = 0. Calculate the linear momentum of the particle w.r.t. the man as a function of time.
[IIT-JEE' 2003]



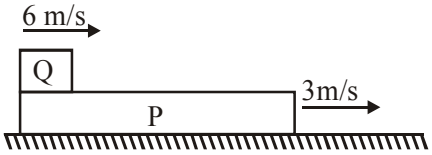
CM0014

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15. A spaceship is moving with constant speed v_0 in gravity free space along +Y-axis suddenly shoots out one third of its part with speed $2v_0$ along + X-axis. Find the speed of the remaining part.

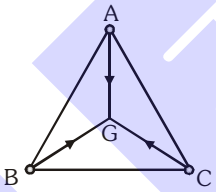
CM0015

16. A plank P and block Q are arranged as shown on a smooth table top. They are given velocities 3 m/s and 6 m/s respectively. The length of plank is 1m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. Find the coefficient of friction between plank P and block Q if mass of plank is double of block).



CM0017

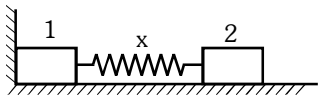
17. Three particles A, B and C of equal mass move with equal speed v along the medians of an equilateral triangle as shown in fig. They collide at the centroid G of the triangle. After the collision, A comes to rest, B retraces its path with the speed v. What is the velocity of C ?



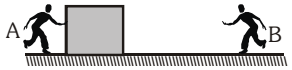
CM0010

18. Two identical buggies move one after the other due to inertia (without friction) with the same velocity v_0. A man of mass m rides the rear buggy. At a certain moment the man jumps into the front buggy with a velocity u relative to his buggy. Knowing that the mass of each buggy is equal to M, find the velocities with which the buggies will move after that. **CM0224**

19. Two bars of masses m_1 and m_2 connected by a weightless spring of stiffness x rest on a smooth horizontal plane. Bar 2 is shifted a small distance x to the left and then released. Find the velocity of the centre of inertia of the system after bar *1* breaks off the wall. **CM0225**



20. Two persons A and B each of mass 100 kg are on a frictionless horizontal surface. Person A at rest is holding a block of mass 50 kg, suddenly pushes the block with some velocity (v) towards the person B approaching at a velocity of 5 m/s. B catches the block & slows down. Now the separation between A and B becomes constant. Find the speed v (in m/s).

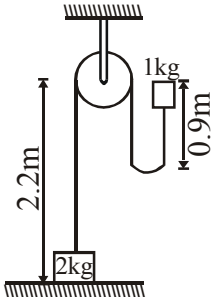


CM0028

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Impulse

21. In the shown figure, the heavy block of mass 2 kg rests on the horizontal surface and the lighter block of mass 1 kg is dropped from a height of 0.9 m. At the instant the string gets taut, find the upward speed (in m/s) of the heavy block. **CM0226**

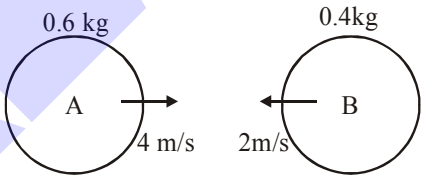


22. Two particles A and B of mass 2m and m respectively are attached to the ends of a light inextensible string of length 4a which passes over a small smooth peg at a height 3a from an inelastic table. The system is released from rest with each particle at a height a from the table. Find
(i) The speed of B when A strikes the table.
(ii) The time that elapses before A first hits the table.
(iii) The time for which A is resting on the table after the first collision & before it is first jerked off.
CM0033

23. A bullet of mass m strikes an obstruction and deviates off at 60° to its original direction. If its speed is also changed from u to v, find the magnitude of the impulse acting on the bullet.
CM0018

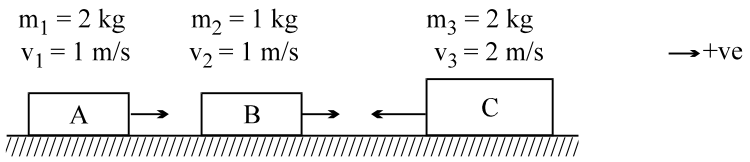
Collision

24. The velocities of two steel balls before impact are shown. If after head on impact the velocity of ball B is observed to be 3 m/s to the right, the coefficient of restitution is



CM0019

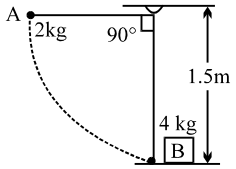
25. Three carts move on a frictionless track with inertias and velocities as shown. The carts collide and stick together after successive collisions.
(i) Find loss of mechanical energy when B & C stick together.
(ii) Find magnitude of impulse experienced by A when it sticks to combined mass (B & C).



CM0020

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26. A sphere A is released from rest in the position shown and strikes the block B which is at rest. If e = 0.75 between A and B and μ_k = 0.5 between B and the support, determine

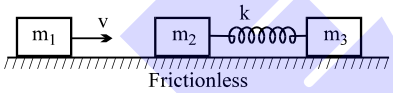


(i) the velocity of A just after the impact
(ii) the maximum displacement of B after the impact. **CM0023**

27. Consider a head-on collision between two particles of masses m_1 and m_2. The initial speeds of the particles are u_1 and u_2 in the same direction. The collision starts at t=0 and the particles interact for a time intercal Δt. During the collision, the speed of the first particle varies as $v(t) = u_1 + \frac{t}{\Delta t}(v_1 - u_1)$
Find the speed of the second particle as a function of time during the collision. **CM0227**

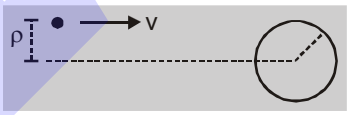
28. A particle of mass 100 g moving at an initial speed u collides with another particle of same mass kept initially at rest. If the total kinetic energy becomes 0.2 J after the collision, what could be the minimum and the maximum value of u. **CM0228**

29. Mass m_1 hits & sticks with m_2 while sliding horizontally with velocity *v* along the common line of centres of the three equal masses ($m_1 = m_2 = m_3 = m$). Initially masses m_2 and m_3 are stationary and the spring is unstretched. Find the



(i) velocities of m_1, m_2 and m_3 immediately after impact.
(ii) maximum kinetic energy of m_3.
(iii) minimum kinetic energy of m_2.
(iv) maximum compression of the spring. **CM0031**

30. A small particle travelling with a velocity v collides elastically with a spherical body of equal mass and of radius r initially kept at rest. The centre of this spherical body is located a distance ρ (<r) away from the direction of motion of the particle (figure). Find the final velocities of the two particles.

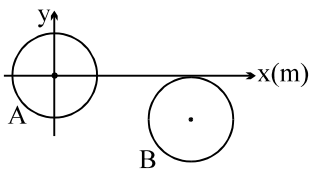


CM0229

31. A particle of mass m_1 collides elastically with a stationary particle of mass m_2 ($m_1 > m_2$). Find the maximum angle through which the striking particle may deviated as a result of the collision.

CM0230

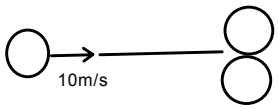
32. Two smooth balls A and B, each of mass m and radius R, have their centres at (0,0,R) and at (5R,–R,R) respectively, in a coordinate system as shown. Ball A, moving along positive x axis, collides with ball B. Just before the collision, speed of ball A is 4 m/s and ball B is stationary. The collision between the balls is elastic. Find Velocity of the ball A just after the collision and impulse of the force exerted by A on B during the collision.



CM0022

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33. A ball with initial speed of 10 m/s collides elastically with two other identical ball whose centres are on a line perpendicular to the initial velocity and which are initially in contact with each other. All the three ball are lying on a smooth horizontal table. The first ball is aimed directly at the contact point of the other two balls All the balls are smooth. Find the velocities of the three balls after the collision.



CM0030

Variable mass and collision of continuous stream

34. Bullets of mass 10 g each are fired from a machine gun at rate of 60 bullets/minute. The muzzle velocity of bullets is 100 m/s. The thrust force due to firing bullets experienced by the person holding the gun stationary is _____.

CM0024

35. A uniform chain of mass M and length L is held vertically in such a way that its lower end just touches the horizontal floor. The chain is released from rest in this position. Any portion that strikes the floor comes to rest. Assuming that the chain does not form a heap on the floor, calculate the force exerted by it on the floor when a length x has reached the floor. **CM0231**

36. During a heavy rain, hailstones of average size 1.0 cm in diameter fall with an average speed of 20 m/s. Suppose 2000 hailstones strike every square meter of a 10 m × 10 m roof perpendicularly in one second and assume that the hailstones do not rebound. Calculate the average force exerted by the falling hailstones on the roof. Density of a hailstone is 900 kg/m^3. **CM0232**

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EXERCISE (JM)

1. This question has Statement I and Statement II. Of the four choices given after the Statements, choose the one that best describes the two Statements. **[JEE Main-2013]**

Statement - I : A point particle of mass m moving with speed v collides with stationary point particle of mass M. If the maximum energy loss possible is given as $f\left(\frac{1}{2}mv^2\right)$ then $f = \left(\frac{m}{M+m}\right)$.

Statement - II : Maximum energy loss occurs when the particles get stuck together as a result of the collision.

(1) Statement–I is true, Statement–II is true, Statement–II is a correct explanation of Statement–I.

(2) Statement–I is true, Statement–II is true, Statement–II is a not correct explanation of Statement–I.

(3) Statement–I is true, Statement–II is false.

(4) Statement–I is false, Statement–II is true **CM0104**

2. A particle of mass m moving in the x direction with speed 2υ is hit by another particle of mass 2m moving in the y direction with speed υ. If the collisions perfectly inelastic , the percentage loss in the energy during the collision is close to : **[JEE Main-2015]**

(1) 56 % (2) 62% (3) 44% (4) 50%

CM0105

3. Distance of the centre of mass of a solid uniform cone from its vertex is z_0. If the radius of its base is R and its height is h then z_0 is equal to :- **[JEE Main-2015]**

(1) $\frac{5h}{8}$ (2) $\frac{3h^2}{8R}$ (3) $\frac{h^2}{4R}$ (4) $\frac{3h}{4}$ **CM0106**

4. It is found that if a neutron suffers an elastic collinear collision with deuterium at rest, fractional loss of its energy is p_d ; while for its similar collision with carbon nucleus at rest, fractional loss of energy is p_c. The values of p_d and p_c are respectively : **[JEE Main-2018]**

(1) (·28, ·89) (2) (0, 0) (3) (0, 1) (4) (·89, ·28)

CM0107

5. In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50 % greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is : **[JEE Main-2018]**

(1) $\sqrt{2}v_0$ (2) $\frac{v_0}{2}$ (3) $\frac{v_0}{\sqrt{2}}$ (4) $\frac{v_0}{4}$ **CM0108**

6. The mass of a hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} hydrogen molecules strike, per second, a fixed wall of area 2 cm^2 at an angle of 45° to the normal, and rebound elastically with a speed of 10^3 m/s, then the pressure on the wall is nearly : **[JEE Main-2018]**

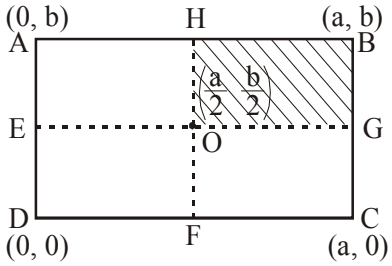
(1) 4.70×10^3 N/m^2 (2) 2.35×10^2 N/m^2 (3) 4.70×10^2 N/m^2 (4) 2.35×10^3 N/m^2

CM0109

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SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

7. A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be :- **[JEE-Main-2019_April]**



(1) $\left(\frac{2a}{3},\frac{2b}{3}\right)$ (2) $\left(\frac{5a}{3},\frac{5b}{3}\right)$ (3) $\left(\frac{3a}{4},\frac{3b}{4}\right)$ (4) $\left(\frac{5a}{12},\frac{5b}{12}\right)$

CM0123

8. A wedge of mass M = 4m lies on a frictionless plane. A particle of mass m approaches the wedge with speed v. There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by :- **[JEE-Main-2019_April]**

(1) $\frac{2v^2}{7g}$ (2) $\frac{v^2}{g}$ (3) $\frac{2v^2}{5g}$ (4) $\frac{v^2}{2g}$

CM0124

9. A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at a speed of 0.70 ms^{-1} with respect to the man. The speed of the man with respect to the surface is : **[JEE-Main-2019_April]**

(1) 0.20 ms^{-1} (2) 0.14 ms^{-1} (3) 0.47 ms^{-1} (4) 0.28 ms^{-1}

CM0125

10. A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is : **[JEE-Main-2020_Jan]**

(1) $\frac{1}{2}$ (2) $\sqrt{\frac{1}{2}}$ (3) $\sqrt{\frac{3}{4}}$ (4) $\sqrt{\frac{3}{2}}$

CM0126

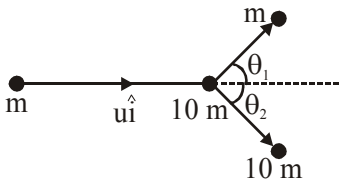
11. Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i}+\hat{j}}{2}\right)$. They collide completely inelastically. The energy lost in the process is : **[JEE-Main-2020_Jan]**

(1) $\frac{3}{4}mu^2$ (2) $\frac{1}{8}mu^2$ (3) $\sqrt{\frac{2}{3}}mu^2$ (4) $\frac{1}{3}mu^2$

CM0127

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12. A particle of mass m is moving along the x-axis with initial velocity $u\hat{i}$. It collides elastically with a particle of mass 10 m at rest and then moves with half its initial kinetic energy (see figure). If $\sin\theta_1 = \sqrt{n}\sin\theta_2$ then value of n is _____. **[JEE-Main-2020_Sep]**



CM0128

13. A block of mass 1.9 kg is at rest at the edge of a table, of height 1 m. A bullet of mass 0.1 kg collides with the block and sticks to it. If the velocity of the bullet is 20 m/s in the horizontal direction just before the collision then the kinetic energy just before the combined system strikes the floor, is [Take $g = 10\ m/s^2$. Assume there is no rotational motion and loss of energy after the collision is negligable.] **[JEE-Main-2020_Sep]**

(1) 21 J (2) 23 J (3) 19 J (4) 20 J **CM0129**

14. Particle A of mass m_1 moving with velocity $(\sqrt{3}\hat{i}+\hat{j})ms^{-1}$ collides with another particle B of mass m_2 which is at rest initially. Let $\vec{V}_1$ and $\vec{V}_2$ be the velocities of particles A and B after collision respectively. If $m_1 = 2m_2$ and after collision $\vec{V}_1 = (\hat{i}+\sqrt{3}\hat{j})ms^{-1}$, the angle between $\vec{V}_1$ and $\vec{V}_2$ is : **[JEE-Main-2020_Sep]**

(1) 60° (2) 15° (3) –45° (4) 105° **CM0130**

15. The centre of mass of a solid hemisphere of radius 8 cm is X cm from the centre of the flat surface. Then value of x is _____ . **[JEE-Main-2020_Sep]**

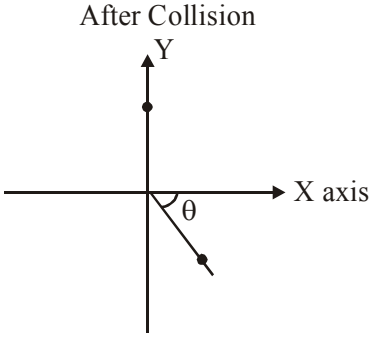
CM0131

16. Two particles having masses 4 g and 16 g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is n : 2. The value of n will be _____. **[JEE-Main-2021_Feb]**

CM0132

17. A ball of mass 10 kg moving with a velocity $10\sqrt{3}\ ms^{-1}$ along X-axis, hits another ball of mass 20 kg which is at rest. After collision, the first ball comes to rest and the second one disintegrates into two equal pieces. One of the pieces starts moving along Y-axis at a speed of 10 m/s. The second piece starts moving at a speed of 20 m/s at an angle θ (degree) with respect to the X-axis.

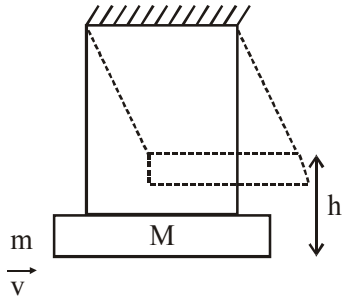
The configuration of pieces after collision is shown in the figure. The value of θ to the nearest integer is _____. **[JEE-Main-2021_March]**



CM0133

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18. A large block of wood of mass M = 5.99 kg is hanging from two long massless cords. A bullet of mass m = 10g is fired into the block and gets embedded in it. The (block + bullet) then swing upwards, their centre of mass rising a vertical distance h = 9.8 cm before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is : (Take g = 9.8 ms^{-2}) **[JEE-Main-2021_March]**



(1) 841.4 m/s (2) 811.4 m/s (3) 831.4 m/s (4) 821.4 m/s

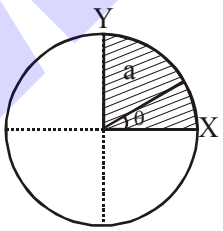
CM0134

19. A rubber ball is released from a height of 5 m above the floor. It bounces back repeatedly, always rising to $\frac{81}{100}$ of the height through which it falls. Find the average speed of the ball. (Take g = 10 ms^{-2}) **[JEE-Main-2021_March]**

(1) 3.0 ms^{-1} (2) 3.50 ms^{-1} (3) 2.0 ms^{-1} (4) 2.50 ms^{-1}

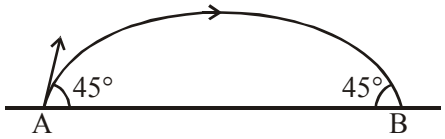
CM0135

20. The disc of mass M with uniform surface mass density σ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position $\frac{x}{3}\frac{a}{\pi}, \frac{x}{3}\frac{a}{\pi}$ where x is _____. (Round off to the Nearest Integer) [a is an area as shown in the figure] **[JEE-Main-2021_March]**



CM0136

21. The projectile motion of a particle of mass 5 g is shown in the figure.



The initial velocity of the particle is $5\sqrt{2}$ ms^{-1} and the air resistance is assumed to be negligible. The magnitude of the change in momentum between the points A and B is x × 10^{-2} $kgms^{-1}$. The value of x, to the nearest integer, is _____.

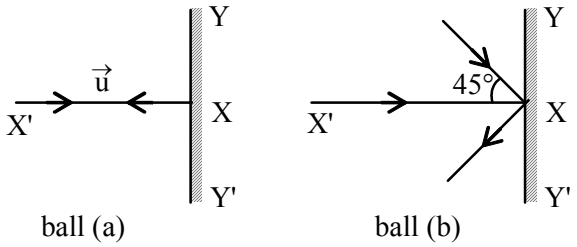
[JEE-Main-2021_March]

CM0137

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22. Two billiard balls of equal mass 30 g strike a rigid wall with same speed of 108 kmph (as shown) but at different angles. If the balls get reflected with the same speed then the ratio of the magnitude of impulses imparted to ball 'a' and ball 'b' by the wall along 'X' direction is :

[JEE-Main-2021_July]



(1) 1 : 1 (2) $\sqrt{2}:1$ (3) 2 : 1 (4) 1 : $\sqrt{2}$

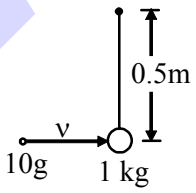
CM0138

23. A body of mass 2 kg moving with a speed of 4 m/s. makes an elastic collision with another body at rest and continues to move in the original direction but with one fourth of its initial speed. The speed of the two body centre of mass is $\frac{x}{10}$ m / s. Then the value of x is_____.

[JEE-Main-2021_July]

CM0139

24. A bullet of 10 g, moving with velocity v, collides head-on with the stationary bob of a pendulum and recoils with velocity 100 m/s. The length of the pendulum is 0.5 m and mass of the bob is 1 kg. The minimum value of v = _____ m/s so that the pendulum describes a circle. (Assume the string to be inextensible and g = 10 m/s^2) **[JEE-Main-2021_Aug]**



CM0140

25. A block moving horizontally on a smooth surface with a speed of 40 ms^{-1} splits into two equal parts. If one of the parts moves at 60 ms^{-1} in the same direction, then the fractional change in the kinetic energy will be x : 4 where x =_____. **[JEE-Main-2021_Aug]**

CM0141

26. A ball of mass 0.15 kg hits the wall with its initial speed of 12 ms^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100 N. calculate the time duration of the contact of ball with the wall. **[JEE-Main-2022_July]**

(1) 0.018 s (2) 0.036 s (3) 0.009 s (4) 0.072 s **CM0233**

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27. A body of mass M at rest explodes into three pieces, in the ratio of masses 1 : 1 : 2. Two smaller pieces fly off perpendicular to each other with velocities of 30 ms^{-1} and 40 ms^{-1} respectively. The velocity of the third piece will be : **[JEE-Main-2022_June]**

(1) 15 ms^{-1} (2) 25 ms^{-1} (3) 35 ms^{-1} (4) 50 ms^{-1} **CM0234**

28. The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\rho = \rho_0\left(1-\frac{x^2}{L^2}\right)$ kg/m and length L (in meter) is $\frac{3L}{\alpha}$ m. The value of α is ……. (where x is the distance form end A) **[JEE-Main-2022_July]**

CM0235

29. A man of 60 kg is running on the road and suddenly jumps into a stationary trolly car of mass 120 kg. Then. the trolly car starts moving with velocity 2 ms^{-1}. The velocity of the running man was _____ ms^{-1}. when he jumps into the car. **[JEE-Main-2022_June]**

CM0236

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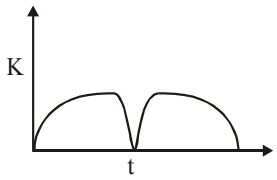
EXERCISE (JA)

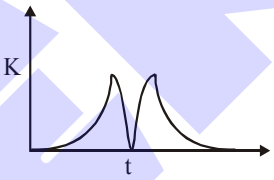
1. A bob of mass m, suspended by a string of length ℓ_1 is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it collides elastically with another bob of mass m suspended by a string of length ℓ_2, which is initially at rest. Both the strings are mass-less and inextensible. If the second bob, after collision acquires the minimum sped required to complete a full circle in the vertical plane, the ratio $\frac{\ell_1}{\ell_2}$ is. **[JEE Advanced-2013]**

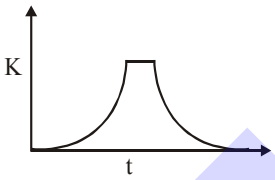
CM0117

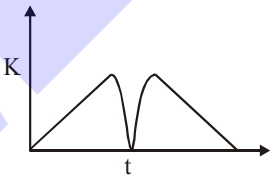
2. A tennis ball is dropped on a horizontal smooth surface. It bounces back to its original position after hitting the surface. The force on the ball during the collision is proportional to the length of compression of the ball. Which one of the following sketches describes the variation of its kinetic energy K with time t most appropriately ? The figures are only illustrative and not to the scale.

[JEE Advanced-2014]

(A) 

(B) 

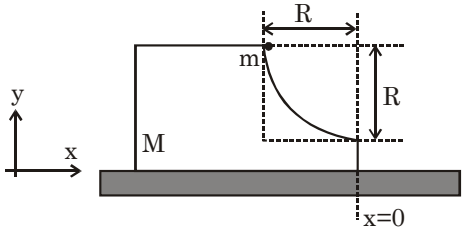
(C) 

(D) 

CM0118

3. A block of mass M has a circular cut with a frictionless surface as shown. The block rests on the horizontal frictionless surface of a fixed table. Initially the right edge of the block is at x = 0, in a *co-ordinate system fixed to the table*. A point mass m is released from rest at the topmost point of the path as shown and it slides down. When the mass loses contact with the block, its position is x and the velocity is v. At that instant, which of the following options is/are **correct** ?

[JEE Advanced-2017]



(A) The x component of displacement of the centre of mass of the block M is : $-\frac{mR}{M+m}$

(B) The position of the point mass is : $x = -\sqrt{2}\frac{mR}{M+m}$

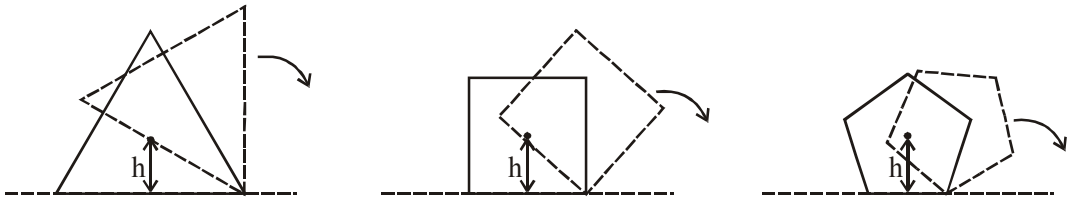
(C) The velocity of the point mass m is : $v = \sqrt{\frac{2gR}{1+\frac{m}{M}}}$

(D) The velocity of the block M is : $V = -\frac{m}{M}\sqrt{2gR}$

CM0119

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4. Consider regular polygons with number of sides n = 3, 4, 5 ..... as shown in the figure. The center of mass of all the polygons is at height h from the ground. They roll on a horizontal surface about the leading vertex without slipping and sliding as depicted. The maximum increase in height of the locus of the center of mass for each polygon is Δ. Then Δ depends on n and h as :**[JEE Advanced-2017]**



(A) $\Delta = h\sin^2\left(\frac{\pi}{n}\right)$ (B) $\Delta = h\sin\left(\frac{2\pi}{n}\right)$ (C) $\Delta = h\left(\frac{1}{\cos\left(\frac{\pi}{n}\right)} - 1\right)$ (D) $\Delta = h\tan^2\left(\frac{\pi}{2n}\right)$

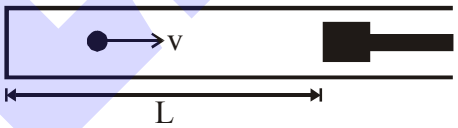
CM0120

5. A solid horizontal surface is covered with a thin layer of oil. A rectangular block of mass m = 0.4kg is at rest on this surface. An impulse of 1.0 N s is applied to the block at time to t = 0 so that it starts moving along the x-axis with a velocity $v(t) = v_0 e^{-t/\tau}$, where v_0 is a constant and τ = 4 s. The displacement of the block, in metres, at $t = \tau$ is..... Take $e^{-1} = 0.37$?

[JEE Advanced-2018]

CM0121

6. A small particle of mass m moving inside a heavy, hollow and straight tube along the tube axis undergoes elastic collision at two ends. The tube has no friction and it is closed at one end by a flat surface while the other end is fitted with a heavy movable flat piston as shown in figure. When the distance of the piston from closed end is $L = L_0$ the particle speed is $v = v_0$. The piston is moved inward at a very low speed V such that $V << \frac{dL}{L} v_0$, where dL is the infinitesimal displacement of the piston. Which of the following statement(s) is/are correct ? **[JEE Advanced-2019]**



(1) The rate at which the particle strikes the piston is v/L

(2) After each collision with the piston, the particle speed increases by 2V

(3) The particle's kinetic energy increases by a factor of 4 when the piston is moved inward from L_0 to $\frac{1}{2}L_0$

(4) If the piston moves inward by dL, the particle speed increases by $2v\frac{dL}{L}$. **CM0122**

Question Stem for Question Nos. 7 and 8

A projectile is thrown from a point O on the ground at an angle 45° from the vertical and with a speed $5\sqrt{2}$ m/s. The projectile at the highest point of its trajectory splits into two equal parts. One part falls vertically down to the ground, 0.5 s after the splitting. The other part, t seconds after the splitting, falls to the ground at a distance x meters from the point O. The acceleration due to gravity $g = 10$ m/s^2.

[JEE Advanced-2021]

7. The value of t is ___ . **CM0142**

8. The value of x is ___ . **CM0142**

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	C	D	C	A	A	D	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	A	B	A	C	C	B	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	A	C	B	D	A	D	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	A	C	C	A	A	A	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	A	D	D	B	B	C	B	D	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	A	A	C	B	C	B	B	C	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	A	B	B	D	C	D	A	A	B	C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	A	C	C	C	A,B,C,D	A,B,D	A,C	A,B,C	A,B,C,D	B,C,D
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	B,C	A,B,D	A,C	A,C	A,B,D	B,C	B	C	D	A-P; B-S; C-Q,R; D-T
Que.	91	92								
Ans.	A-Q; B-S; C-P	A-R; B-T; C-T; D-S								

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	C	C	A	C	B	A	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	B	B	B	B	A	C	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	C	D	B,C	A,B	B,C	A,B,C,D	A	C
Que.	31	32								
Ans.	A-Q; B-Q; C-P,R; D-Q,S	A-Q; B-P,Q; C-R; D-S								

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EXERCISE # S

1. (1/7, 23/14) **2.** $L(\sqrt{2}+1)/3$ **3.** 4R from O **4.** $\frac{4\left(R_1^2+R_1R_2+R_2^2\right)}{3\pi\left(R_1+R_2\right)}$ **above the centre**

5. $\frac{m(R-r)}{M+m}$, $m\sqrt{\frac{2g(R-r)}{M(M+m)}}$ **6.** $\frac{L}{3}$ **7.** $\frac{R}{4}$ **8.** $\tilde{E} = \mu(v_1^2 + v_2^2)/2$, where $\mu = m_1m_2/(m_1 + m_2)$

9. (a) $F_{fr} = mg\,[\sin\alpha + (\omega^2\ell/g)\cos\alpha] = 6N$ (b) $\omega << \sqrt{g(k-\tan\alpha)/\ell(1+k\tan\alpha)} = 2$ rad/s. **10.** $\frac{2mv^2}{3\ell}$

11. (a) $a_c = \frac{F}{2m}$, (b) $x_c = \frac{Ft^2}{4m}$; (c) $x_A = \frac{1}{2}\left(\frac{Ft^2}{2m} - x_0\right)$ and $x_B = \frac{1}{2}\left(\frac{Ft^2}{2m} + x_0\right)$ **12.** 150 kg

13. (i) 4 m/s, (ii) 24 J **14.** $\vec{P}_{PM} = m\,\vec{v}_{PM} = -mv_2\sin\omega t\,\hat{i} + m(v_2\cos\omega t - v_1)\,\hat{j}$ **15.** $\frac{\sqrt{13}}{2}v_0$

16. 0.3 **17.** $\vec{v}_C = -\vec{v}_B$ **18.** $v_{rear} = v_0 - \frac{m}{M+m}u$; $v_{form} = v_0 + \frac{mM}{(M+m)^2}u$ **19.** $v_C = x\sqrt{xm_2}/(m_1 + m_2)$

20. Ans. 4 **21. 2** **22.** (i) $\sqrt{\frac{2ag}{3}}$ (ii) $\frac{3v}{g}$ (iii) $\frac{2v}{g}$ **23.** $m \times \sqrt{u^2 - uv + v^2}$

24. $\frac{7}{18}$ **25.** (i) 3 J, (ii) $\frac{12}{5}$ N-s **26.** (i) $v_A = \sqrt{g/12}$ m/s, (ii) $S_{max} = 49/48$ m **27.** $u_2 - \frac{m_1}{m_2}\frac{t}{\Delta t}(v_1 - u_1)$

28. 2 m/s, $2\sqrt{2}$ m/s **29.** (i) v/2, v/2, 0; (ii) $2mv^2/9$; (iii) $mv^2/72$; (d) $x = \sqrt{m/6k}\,v$

30. The small particle goes along the tangent with a speed of $v\rho/r$ and the spherical body goes perpendicular to the smaller particle with a speed of $\frac{v}{r}\sqrt{r^2 - \rho^2}$

31. $\sin\theta_{max} = m_2/m_1$ **32.** $(\hat{i} + \sqrt{3}\hat{j})$ m/s, $(3m\hat{i} - \sqrt{3}m\hat{j})$ kg-m/s

33. –2m/s, 6.93 m/s $\angle$ 30° **34.** 1 N **35. 3 Mgx/L** **36. 1900 N**

EXERCISE # JM

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	4	4	1	4	4	3	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	10	1	4	3	1	30	3	4	4
Que.	21	22	23	24	25	26	27	28	29	
Ans.	5	2	25	400	1	2	2	8	6	

EXERCISE # JA

Que.	1	2	3	4	5	6	7	8
Ans.	5	B	A,C	C	6.3 (6.29, 6.31)	2,3	0.50	7.50

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